

Manuscripts considered — gbm-mtap-cdkn2a-idhwt-subtotal-c4bq

Master inventory: every paper reviewed in this case

Generated 2026-05-26

Manuscripts considered — gbm-mtap-cdkn2a-idhwt-subtotal-c4b

Master inventory: 37 clinical (34 included, 3 considered & excluded) + 42 pre-clinical (35 included, 7 considered & excluded) + 27 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Sanai N/Sanai N (2026) — ClinicalTrials.gov record (Ivy Brain Tumor Center)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BGB-58067 (brain-penetrant MTA-cooperative PRMT5 inhibitor)
- **Indication / model:** Newly diagnosed MTAP-deleted IDH-wildtype glioblastoma (1L); MTAP loss by IHC on resection tumor; stratified by MGMT methylation
- **Design:** Phase 0/2 dual-arm; Arm A unmethylated-MGMT (BGB-58067 + RT), Arm B methylated-MGMT (BGB-58067 + RT + TMZ)
- **n:** 78
- **Effect size:** — — Phase 0: tumor SDMA reduction $\geq 50\%$ or H-score ≤ 70 ; Phase 2: 12-month OS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (MountainTAP-5)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Navlimetostat (BMS-986504) monotherapy or combination
- **Indication / model:** Advanced/metastatic solid tumors with homozygous MTAP deletion (GBM explicitly listed) (2L+); MTAP homozygous deletion
- **Design:** Phase 2 open-label monotherapy and combinations across MTAP-deleted tumor types
- **Effect size:** — — ORR per RECIST 1.1 / RANO v2 / mRECIST 1.1
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (IDEAYA)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IDE892 (next-generation MAT2A inhibitor)
- **Indication / model:** MTAP-deleted advanced solid tumors (NSCLC, GEJ, urothelial, mesothelioma; no GBM cohort listed at start) (2L+); MTAP homozygous deletion
- **Design:** Phase 1 monotherapy and combination dose-escalation
- **Effect size:** — — DLT, RP2D, safety
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Tango Therapeutics)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TNG462 (peripheral-only PRMT5 inhibitor)
- **Indication / model:** MTAP-deleted advanced solid tumors EXCLUDING primary CNS (2L+); MTAP biallelic deletion (NGS or IHC)
- **Design:** Phase 1/2 dose-escalation and expansion
- **n:** 225
- **Effect size:** — — DLT, RP2D, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Okada H/Okada H (2026) — ClinicalTrials.gov record (UCSF, E-SYNC)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** E-SYNC: autologous anti-EGFRvIII synNotch receptor inducing anti-EphA2 / IL-13Ra2 CAR T cells
- **Indication / model:** EGFRvIII+ glioblastoma (Cohort 1 newly diagnosed unmethylated MGMT; Cohort 2 recurrent) (1L or 2L); EGFRvIII confirmed by CLIA NGS panel; unmethylated MGMT (Cohort 1)
- **Design:** Phase 1 dose-escalation, two sequential cohorts
- **n:** 20
- **Effect size:** — — Treatment-emergent AEs
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Bagley SJ/O'Rourke DM (2026) — ClinicalTrials.gov record (Penn)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CART-EGFR-IL13Ra2 (dual-targeted CAR T cells; Bagley/Penn next-generation construct)
- **Indication / model:** Recurrent glioblastoma with EGFR amplification (wild-type) (2L+); EGFR amplification (no vIII requirement); IL13Ra2 is the second CAR target but not gating for tumor selection
- **Design:** Phase 1b three-arm dose-schedule comparison
- **Effect size:** — — Treatment-emergent AEs and treatment-limiting toxicities
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Global Coalition for Adaptive Research)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Current arms: VT1021, troriluzole, ADI-PEG 20, AZD1390, tinostamustine (regorafenib/paxalisib/VAL-083 closed)
- **Indication / model:** Newly diagnosed and recurrent IDH-wildtype glioblastoma; adaptive platform (1L and 2L+ (separate strata)); Both MGMT-methylated and unmethylated enrolled; arm assignment by biomarker subgroup
- **Design:** Bayesian adaptive platform; multiple investigational arms vs. shared control
- **Effect size:** — — OS (primary)
- **Case fit:** partial

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (NovoCure, TRIDENT EF-32)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Optune (TTFields 200 kHz) concomitant with RT + TMZ, then continued with maintenance TMZ
- **Indication / model:** Newly diagnosed glioblastoma (1L); Both MGMT-methylated and unmethylated enrolled (stratification at randomization)
- **Design:** Randomized open-label pivotal phase 3
- **Effect size:** — — OS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (DCVax-L expanded access)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DCVax-L compassionate-use access
- **Indication / model:** Newly diagnosed GBM with manufacturable DCVax-L lysate (1L); None required (autologous lysate)
- **Design:** Expanded-access protocol
- **Effect size:** — — —
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Ahluwalia MS/Ahluwalia MS (2026) — ClinicalTrials.gov record (SURVIVE; MimiVax)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SurVaxM (survivin peptide vaccine) + adjuvant TMZ
- **Indication / model:** Newly diagnosed glioblastoma (1L (adjuvant)); MGMT status required at enrollment (both methylated and unmethylated eligible)
- **Design:** Randomized open-label phase 2
- **Effect size:** — — OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Istari Oncology / Duke)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Lerapolturev (PVSRIPO, oncolytic poliovirus) intratumoral
- **Indication / model:** Recurrent glioblastoma (2L+); None required
- **Design:** Randomized phase 2 lerapolturev vs lomustine
- **n:** 92
- **Effect size:** — — OS
- **Case fit:** weak

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Dana-Farber)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** B7-H3.CD28Z CAR T cells
- **Indication / model:** B7-H3-expressing CNS neoplasms (peds + young adult) (2L+); B7-H3 expression
- **Design:** Phase 1/1b dose-escalation
- **Effect size:** — — Safety
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Beijing Biotech)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Dual-targeting CAR-NK cells
- **Indication / model:** Recurrent/progressive glioblastoma (2L+); Dual-target: B7-H3 + IL13Ra2 or EGFR/EGFRvIII
- **Design:** Phase 1 dose-escalation of dual-target CAR-NK
- **Effect size:** — — Safety, feasibility
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2026) — ClinicalTrials.gov record (Johns Hopkins SKCCC)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Carmustine wafer (Gliadel) + retifanlimab (PD-1) + RT ± TMZ
- **Indication / model:** Newly diagnosed glioblastoma (1L); Not stratified by MGMT
- **Design:** Phase 1 with several escalation arms (with and without TMZ)
- **Effect size:** — — Safety, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Leddon J/Leddon J (2026) — ClinicalTrials.gov record (Cincinnati)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pembrolizumab + navigated focused ultrasound (FUS-mediated BBB opening)
- **Indication / model:** Recurrent WHO grade 4 IDH-wildtype GBM with MMR deficiency (2L+); MMR deficient (dMMR)
- **Design:** Single-arm phase 1 pilot
- **n:** 8
- **Effect size:** — — Safety, feasibility
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Rodon J/Hyman DM (2025) — Cancer Discovery

Reference:PMID 39282709 · Type: clinical · Inclusion: included

- **Intervention:** Anvumetostat (AMG 193) — translational + clinical companion
- **Indication / model:** MTAP-deleted advanced solid tumors (2L+); Combined preclinical + early clinical translational characterization; patient cohorts overlap with the AnnOnc 2024 phase 1 paper.
- **Design:** Translational + phase 1 combined report
- **n:** 80
- **Effect size:** Robust antitumor activity in MTAP-null CDX/PDX models; clinical ORR and ctDNA clearance corroborating — Mechanism + clinical activity in MTAP-deleted cancers
- **Toxicities:** Safety detailed in the AnnOnc 2024 phase 1 paper; same dataset. Translational sections in this paper carry preclinical hematologic-sparing data (no impact on normal hematopoietic lineages in xenografts).
- **Case fit:** cross_tumor_only
- **Notes:** Companion translational paper to the AnnOnc 2024 phase 1. Cancer Discovery is the higher-impact venue and carries the preclinical + mechanism-of-action data; both rows together comprise the foundational anvumetostat evidence base. Toxicities captured on the AnnOnc 2024 row to avoid double-counting; this row's value is the mechanism context.

Desjardins A/Bigner DD (2025) — Clinical Cancer Research

Reference:PMID 40313264 · Type: clinical · Inclusion: included

- **Intervention:** Lerapolturev (PVSRIPO) — phase 2 update
- **Indication / model:** Recurrent WHO grade IV malignant glioma (2L+); 121 patients pooled across two phase 2 cohorts at Duke and partner sites. KPS ≥ 70 , single CED-accessible lesion.
- **Design:** Single-arm multicenter phase 2
- **n:** 121
- **Effect size:** 7.0 months — OS
- **Toxicities:** peritumoral inflammation / edema; peritumoral inflammation $G \geq 3$
- **Case fit:** weak
- **Notes:** Phase 2 update — mOS lower than the 2018 phase 1 (7.0 vs 12.5 mo) but the 24-mo OS plateau (~19%) confirms the long-tail benefit pattern. Randomized phase 2 vs lomustine (NCT06177964) is the next maturation.

Cottrell KM/Maxwell JP (2025) — Journal of Medicinal Chemistry

Reference:PMID 40035511 · Type: preclinical · Inclusion: included

- **Intervention:** TNG462 / vopimetostat — peripheral MTA-cooperative PRMT5 inhibitor
- **Indication / model:** HAP1 MTAP-null and MTAP-WT isogenic pair; CDX and PDX panel (pancreatic, NSCLC, mesothelioma, bladder); rodent and dog PK
- **Design:** Same MTA-cooperative mechanism as TNG908 but optimized for potency and peripheral PK ($T_{1/2} > 24$ h supporting QD dosing). Brain penetration is intentionally low — TNG462 is the peripheral follow-on after TNG908 disclosed CNS limitations.
- **n:** n=8-10 mice/arm in efficacy studies; multiple PDX lineages
- **Effect size:** strong — TNG462 has biochemical K_i estimated at ≤ 300 fM and 45-fold selectivity for MTAP-null cells; produces regressions in pancreatic, NSCLC, mesothelioma, and bladder PDX models. Combination with

KRAS G12C and EGFR inhibitors gave deeper regressions than either agent alone.

- **Variance:** vs MTAP-WT isogenic and vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Explicitly NOT a CNS asset — the TNG462 clinical protocol excludes primary CNS tumors. Relevant to the case only as the combination partner in the Servier S095035 + TNG462 trial (NCT06188702), where the small-molecule MAT2A partner may carry the activity into GBM-eligible cohorts.

Rodon J/Hyman DM (2025) — Cancer Discovery

Reference:PMID 39282709 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** AMG 193 / anvetmetostat — MTA-cooperative PRMT5 inhibitor (preclinical)
- **Indication / model:** MTAP-null CDX and PDX panel across multiple lineages (lung, pancreatic, biliary); isogenic MTAP knockout pairs; normal hematopoietic lineage assays
- **Design:** MTA-cooperative PRMT5 binding. The selectivity advantage over first-generation PRMT5 inhibitors (GSK3326595, JNJ-64619178) translates to a preserved normal-hematopoiesis profile — the hematologic toxicity that capped the dose of earlier compounds is largely absent.
- **n:** Multiple PDX/CDX models, n=8-10 mice/arm typical
- **Effect size:** strong — Robust antitumor activity across MTAP-null xenografts; no impact on normal hematopoietic cell lineages at efficacious doses. Synergy with chemotherapy and with the KRAS G12C inhibitor sotorasib in combination studies. SDMA reduction tracks tumor response and is the proposed pharmacodynamic biomarker.
- **Variance:** vs MTAP-WT isogenic; vehicle; first-generation non-selective PRMT5 inhibitor comparators in some assays; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No GBM-specific in-vivo arm. AMG 193 is not optimized for CNS penetration — the Part 1m glioma cohort in MTAPESTRY 101 is the empirical test. The preclinical work anchors the broader MTA-cooperative class safety profile, which informs why BGB-58067 (brain-penetrant successor with the same MTA-cooperative mechanism) is the lead asset for this patient.

Jiang Z/Li X (2025) — Cell Death Discovery

Reference:PMID 40450009 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRMT5 + MAT2A combined inhibition in MTAP-null glioma (dedicated glioma study)
- **Indication / model:** MTAP-null vs MTAP-WT glioma cell lines (U87, U251, primary patient-derived); orthotopic mouse glioma xenograft
- **Design:** Combined PRMT5 and MAT2A inhibition in MTAP-deficient glioma produces deeper suppression of SDMA than either agent alone — MAT2A inhibition lowers SAM so MTA wins the binding competition, while PRMT5 inhibitor occupies the active site directly. Two complementary nodes squeeze the same axis from both sides.
- **n:** Multiple cell lines; in-vivo arm n=6-8 mice/group
- **Effect size:** strong — Combined PRMT5 + MAT2A inhibition gave strong synthetic lethality in MTAP-homozygous-deficient glioma cells; the combination outperformed either agent alone in vitro and extended survival in orthotopic models. This is the closest published preclinical evidence to the Servier S095035 + TNG462 combination rationale.

- **Variance:** vs MTAP-WT isogenic; vehicle; single-agent comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Newer paper (May 2025) — replication outside the Soochow group not yet published. In-vivo arm uses xenografts in immunocompromised mice, not syngeneic immunocompetent models, so the immune-context biology is not addressed. The direct relevance to the patient is that this is the only dedicated PRMT5 + MAT2A combination study in MTAP-null glioma models published to date.

Onishi S/Banasavadi-Siddegowda YK (2025) — Research Square (preprint; Cell Death & Disease in press)

Reference: [PMID 39989968](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRMT5 inhibition + temozolomide — preclinical synergy in glioma
- **Indication / model:** Patient-derived GBM neurospheres; orthotopic mouse xenograft
- **Design:** PRMT5 methylates components of the Fanconi anemia / homologous recombination DNA-repair machinery. PRMT5 inhibition impairs HR-mediated repair of TMZ-induced O6-methylguanine and double-strand-break intermediates, increasing TMZ cytotoxicity in MGMT-low or MGMT-methylated tumors and partially rescuing benefit even in MGMT-unmethylated lines.
- **n:** Multiple neurosphere lines; n=8-10 mice/arm in vivo
- **Effect size:** strong — LLY-283 + TMZ in GBM neurospheres produced increased apoptosis (caspase 3/7), more γH2AX foci, and more comet-tail DNA damage than TMZ alone. In orthotopic xenografts, the combination significantly prolonged survival over either single agent. The mechanism is HR repair failure rather than MGMT-pathway interaction.
- **Variance:** vs TMZ alone; PRMT5i alone; vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Research Square preprint as of data cut; peer-reviewed version pending in Cell Death & Disease. The HR-repair mechanism is the part most likely to generalize across MGMT subgroups; the survival signal will need replication beyond the originating lab. Directly relevant to the patient's pending MGMT result — if unmethylated, PRMT5i + TMZ becomes a mechanism-supported combination even when MGMT methylation is not driving TMZ benefit on its own.

Smith K/Wilcoxon K (2025) — Cancer Research / AACR 2025 abstract 2996

Reference: [doi:10.1158/1538-7445.AM2025-2996](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** TNG462 + CDK4/6 inhibitor in MTAP-null preclinical models (AACR 2025)
- **Indication / model:** MTAP-null CDX and PDX panel (NSCLC, pancreatic, bladder); TNG462 ± palbociclib or abemaciclib combination dosing
- **Design:** TNG462 inhibits PRMT5 selectively in MTAP-null cells; CDK4/6 inhibitors block Rb phosphorylation. The combination converges on G1/S blockade through two distinct nodes — PRMT5-dependent splicing of cell-cycle genes plus direct cyclin-D/CDK4/6 inhibition. Particularly attractive in MTAP+CDKN2A co-deleted tumors (the 9p21 co-deletion phenotype the patient carries).
- **n:** Multiple PDX lines; n=8-10 mice/arm
- **Effect size:** strong — TNG462 + CDK4/6 inhibitor produced deeper and more durable regressions than either agent alone across MTAP-null PDX models. The combination was tolerated and showed clear additivity in MTAP+CDKN2A co-deleted lines — the sponsor explicitly nominated this as a development path for GBM

(using a CNS-penetrant PRMT5i in place of TNG462, since TNG462 itself does not cross the BBB).

- **Variance:** vs Either agent alone; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Conference abstract — full peer-reviewed manuscript pending. No GBM cohort in the actual preclinical models tested; the GBM development path is the sponsor's stated intent, not yet executed. Still the most directly relevant preclinical combination data for this patient's MTAP+CDKN2A co-deletion phenotype.

Erden M/Various (2025) — MDPI Encyclopedia / R Discovery preprint

Reference: — · **Type:** preclinical · **Inclusion:** excluded — Review article, not primary preclinical data. The primary papers it synthesizes are already captured in the included rows (Kryukov 2016, Mavrakis 2016, Marjon 2016, Sachamitr 2021, Engstrom 2024, Smith 2023, Kalev 2021).

- **Intervention:** PRMT5 synthetic lethality in MTAP-deficient GBM — review
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Review article, not primary preclinical data. The primary papers it synthesizes are already captured in the included rows (Kryukov 2016, Mavrakis 2016, Marjon 2016, Sachamitr 2021, Engstrom 2024, Smith 2023, Kalev 2021).

—/— (2025) — ClinicalTrials.gov record (Tango Therapeutics)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TNG908 (brain-penetrant PRMT5 inhibitor)
- **Indication / model:** MTAP-deleted advanced solid tumors including relapsed/refractory GBM (dedicated GBM expansion arm) (2L+); MTAP biallelic deletion (NGS or IHC)
- **Design:** Phase 1/2 multi-arm; the GBM arm was a dedicated Phase 2 expansion at the RP2D
- **n:** 110
- **Effect size:** — — DLT, ORR (RANO for the GBM arm)
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wen PY/Wen PY (2025) — ClinicalTrials.gov record (INSIGHt — Dana-Farber)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Standard TMZ control vs. neratinib + TMZ vs. abemaciclib + TMZ (closed to accrual) vs. CC-115 (closed) vs. QBS10072S
- **Indication / model:** Newly diagnosed IDH-wildtype GBM, MGMT unmethylated (1L (adjuvant)); MGMT unmethylated; biomarker subgroups by WES/CN drive Bayesian adaptive randomization
- **Design:** Phase 2 Bayesian adaptive randomization platform; biomarker-stratified arms
- **n:** 460
- **Effect size:** — — OS at 2 years
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record / Nature 2023 (Ling)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CAN-3110 (rQNestin34.5v.2, oncolytic HSV-1)
- **Indication / model:** Recurrent malignant glioma (2L+); None required
- **Design:** Phase 1 dose-escalation; intratumoral injection
- **Effect size:** Single-dose median OS 11.6 mo in nestin-expressing recurrent HGG cohort — Safety, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record (Zhejiang University)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** B7-H3 CAR T cells
- **Indication / model:** Recurrent or refractory glioblastoma (2L+); B7-H3 expression
- **Design:** Phase 1 pilot dose-escalation
- **Effect size:** — — Safety
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2025) — ClinicalTrials.gov record (Binhui)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** OH2 oncolytic HSV-2
- **Indication / model:** CNS tumors (recurrent malignant glioma) (2L+); None required
- **Design:** Phase 1/2a
- **Effect size:** — — Safety, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wen PY/Wen PY (2024) — Neuro-Oncology (conference abstracts) / Dana-Farber trial record

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** Abemaciclib (recurrent GBM, CDKN2A-loss / RB-intact)
- **Indication / model:** Recurrent GBM or IDH-mutant grade 4 astrocytoma at first relapse, CDKN2A/B-deleted or CDK4/6-amplified with intact RB (2L); Adults with first-relapse GBM, KPS ≥ 60 , CDKN2A/B homozygous deletion or CDK4/6 amplification by NGS, intact RB1 protein expression by IHC. Phase 0 paired-surgical cohort plus phase 2 non-surgical cohort.
- **Design:** Phase 0/2 single-arm with paired surgical PK
- **n:** 42
- **Effect size:** 9.7 % — PFS6, intratumoral abemaciclib concentration
- **Variance:** 95% CI 2.5–22.9
- **Toxicities:** diarrhea; neutropenia ≥ 3 ; fatigue
- **Case fit:** partial
- **Notes:** NCT02981940; full peer-reviewed manuscript pending. PFS6 $\sim 9.7\%$ defines the activity ceiling for

abemaciclib monotherapy in recurrent CDKN2A-loss GBM — informs how aggressively to chase CDK4/6i at recurrence vs. layering in combinations (e.g. + ERKi LY3214996, the CTNI-58 successor arm). The paired-surgical arm confirmed intratumoral drug concentrations sufficient for CDK4/6 inhibition, which the palbociclib failure (Taal 2018) had not achieved. Safety data not fully published; full per-term toxicities will become available with the planned manuscript — capture them then.

Rodon J/Hyman DM (2024) — Annals of Oncology

Reference:doi:10.1016/j.annonc.2024.06.005 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Anvumetostat (AMG 193) — first-in-human phase 1
- **Indication / model:** Advanced MTAP-deleted solid tumors (lung, pancreatic, biliary, glioma — pan-tumor) (2L+); Patients with MTAP-deleted solid tumors. Central MTAP IHC mandated after early enrollment showed four-of-six false-positive CDKN2A-only-call patients lacked MTAP loss. ECOG 0-1.
- **Design:** Open-label phase 1 dose-escalation (MTAPESTRY 101)
- **n:** 80
- **Effect size:** ORR ~11% across MTAP-null escalation cohort; durable PRs in NSCLC, pancreatic, biliary % — DLT, RP2D, preliminary ORR, SDMA target engagement
- **Toxicities:** any TRAE G $\geq$ 3 (n=80, 13.8%); treatment discontinuation due to TRAE (n=80, 2.5%); fatigue; nausea; anemia G $\geq$ 3
- **Case fit:** cross_tumor_only
- **Notes:** MTAPESTRY 101 first-in-human dose-exploration readout. Anvumetostat is the most clinically-mature MTA-cooperative PRMT5 inhibitor and the safety/efficacy profile here anchors the rationale for BGB-58067 (brain-penetrant successor) and the broader MTA-cooperative class. The four-of-six false-positive CDKN2A-only-call enrollment finding is why central MTAP IHC is mandatory across the class — directly relevant to this patient's diagnostic gate plan.

Hamilton EP/— (2024) — Tango Therapeutics press release / ESMO 2024

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** TNG908 (MTA-cooperative brain-penetrant PRMT5 inhibitor — discontinued)
- **Indication / model:** MTAP-deleted advanced solid tumors including GBM (dedicated GBM expansion arm) (2L+); 33 GBM patients + 70 non-CNS patients across 24 histologies, all MTAP-deleted. Data cutoff 20 Oct 2024.
- **Design:** Phase 1/2 open-label dose-escalation and expansion
- **n:** 103
- **Effect size:** 0% ORR in 23 GBM patients at active doses; clinical activity in non-CNS (NSCLC, pancreatic) % — DLT, RP2D, ORR (by tumor type)
- **Toxicities:** any TRAE G $\geq$ 3
- **Case fit:** weak
- **Notes:** Sponsor announcement Nov 2024 (Tango Therapeutics 8-K and ESMO presentation): TNG908 discontinued in GBM because CSF samples confirmed inadequate brain exposure, not because of toxicity. Decision-relevant context for this case: it explains why BGB-58067 — the BeiGene brain-penetrant successor with explicit CNS exposure modeling in the IND — became the new lead asset for MTAP-deleted GBM, and why TNG462 (Tango's peripheral-only follow-on) excludes primary CNS tumors. Patient was eligible for TNG908 GBM expansion when it was open; now must wait for BGB-58067 (NCT07485049) or navlimetostat (NCT07492680, MountainTAP-5). Safety reporting limited to the press-release abstract — PubMed/PMC

fallback yielded no peer-reviewed paper; the AE data is held in sponsor presentations only. Mark for backfill when peer review publishes.

Bagley SJ/O'Rourke DM (2024) — Nature Medicine

Reference: [PMID 38497615](#) · Type: clinical · Inclusion: included

- **Intervention:** CART-EGFR-IL13Ra2 (bivalent intrathecal Penn construct)
- **Indication / model:** Recurrent EGFR-amplified glioblastoma (2L+); 6 adults with recurrent GBM, EGFR amplification (no vIII required), measurable disease. Intrathecal delivery via Ommaya reservoir.
- **Design:** Phase 1 dose-escalation interim analysis, single-center (Penn)
- **n:** 6
- **Effect size:** 6/6 patients showed tumor volume reduction within 2 days of infusion; no patient met RANO objective response criteria; responses generally transient — — Safety, radiographic tumor reduction
- **Toxicities:** neurotoxicity (ICANS-like) (n=6, 100%); neurotoxicity G $\geq$ 3; CRS; headache (n=6, 100%); fatigue (n=6, 100%)
- **Case fit:** weak
- **Notes:** Phase 1 interim with n=6. Dramatic radiographic responses within 2 days of intrathecal infusion are hypothesis-generating but did not meet formal RANO objective-response criteria. The dual-target design (EGFR + IL13Ra2) was the lab's response to single-antigen escape seen in the 2017 STM paper. Patient is currently newly diagnosed so this is recurrence-relevant only; conditional on EGFR amplification on the comprehensive NGS panel. The intratumoral hemorrhage history is a relative caution given the neurotoxicity profile.

Choi BD/Maus MV (2024) — N Engl J Med

Reference: [PMID 38477966](#) · Type: clinical · Inclusion: included

- **Intervention:** CARv3-TEAM-E T cells (intraventricular MGH construct)
- **Indication / model:** Recurrent EGFR-expressing glioblastoma (2L+); 3 adults with recurrent GBM expressing EGFR (not requiring vIII); intraventricular delivery via Ommaya. Single-center MGH.
- **Design:** Phase 1 first-in-human, single-center
- **n:** 3
- **Effect size:** 3/3 had rapid radiographic regression within days of infusion; 2/3 progressed within 1-2 months; 1/3 had durable response sustained at 150 days — — Safety, radiographic response
- **Toxicities:** any TRAE G $\geq$ 3 (n=3, 0%); fever; headache; CRS G $\geq$ 2 (n=3, 0%); ICANS G $\geq$ 3 (n=3, 0%)
- **Case fit:** weak
- **Notes:** n=3 single-center; dramatic radiographic responses anchor the rationale for the CAR-T-plus-T-cell-engager combination class. Patient is not enrollable here (recurrent disease at MGH) but the trial is part of the recurrence-conversation menu if EGFR expression confirms. The intratumoral hemorrhage history is a relative caution — intraventricular CAR-T plus rapid tumor lysis raised intracranial edema concerns in this cohort, all resolved with steroids.

Cottrell KM/Maxwell JP (2024) — Journal of Medicinal Chemistry

Reference: [PMID 38595098](#) · Type: preclinical · Inclusion: included

- **Intervention:** TNG908 — brain-penetrant MTA-cooperative PRMT5 inhibitor (preclinical)
- **Indication / model:** MTAP-null cell line panel including U87MG MTAP-null derivatives; orthotopic GBM mouse

- xenograft; non-human primate brain PK; rodent PK with P-gp/BCRP transporter assays
- **Design:** TNG908 is an MTA-cooperative PRMT5 inhibitor engineered to avoid P-gp and BCRP efflux. Selectivity for MTAP-null cells comes from the MTA-bound ternary complex; brain penetrance comes from passive permeability with no active efflux.
- **n:** n=8-10 mice/arm in the orthotopic survival study; multiple cell lines for in-vitro selectivity
- **Effect size:** strong — TNG908 was 15-fold more potent in MTAP-null vs MTAP-WT lines and produced near tumor stasis with ~3-fold median survival extension in an aggressive orthotopic GBM xenograft. Brain Kp,uu approached unity in rodent and was preserved in non-human primate — the design metric for CNS exposure was met preclinically.
- **Variance:** vs MTAP-WT cell lines; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Preclinical CNS exposure looked excellent but the clinical CSF readout in the GBM expansion arm came in low — Tango disclosed inadequate brain exposure as the proximate failure mechanism in Nov 2024. A reminder that rodent Kp,uu does not always predict human CSF PK. Superseded clinically by BGB-58067; the preclinical mechanism work still anchors the class rationale.

Hartmann C/Schiff D (2024) — Clinical Cancer Research

Reference: [PMID 38718141](#) · **Type:** preclinical · **Inclusion:** excluded — Population is IDH-mutant glioma, not the patient's IDH-wildtype GBM. The CDKN2A loss × CDK4/6i sensitivity logic is the same but the tumor biology is materially different; the patient's IDH-WT axis is covered by Li 2017 and Parrish 2015. Logged for audit trail.

- **Intervention:** CDKN2A/B homozygous deletion sensitizes IDH-mutant glioma to CDK4/6 inhibition
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Population is IDH-mutant glioma, not the patient's IDH-wildtype GBM. The CDKN2A loss × CDK4/6i sensitivity logic is the same but the tumor biology is materially different; the patient's IDH-WT axis is covered by Li 2017 and Parrish 2015. Logged for audit trail.

Karim L/Bashir T (2024) — Nature Communications (and AACR 2023 abstract)

Reference: [doi:10.1038/s41467-024-55316-5](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** IDE397 — second-generation MAT2A inhibitor (preclinical and early clinical)
- **Indication / model:** MTAP-null CDX and PDX panel (NSCLC, urothelial, esophageal, mesothelioma); biochemical and cellular MAT2A inhibition; SAM depletion as PD readout
- **Design:** IDE397 is an allosteric MAT2A inhibitor that blocks SAM synthesis in MTAP-null cells. Lower SAM (combined with elevated MTA from MTAP loss) collapses residual PRMT5 activity past the lethal threshold. Designed for better DMPK properties than first-generation AG-270.
- **n:** Multiple PDX lines, n=8-10 mice/arm
- **Effect size:** strong — IDE397 produced regressions across MTAP-null PDX lineages (urothelial, NSCLC dominant) and reduced plasma SAM in a dose-dependent manner that tracks tumor response. Clinical phase 1 expansion at 30 mg QD reported 33% ORR in evaluable MTAP-null urothelial / NSCLC patients (8 PR + 1 CR in n=27).
- **Variance:** vs MTAP-WT isogenic and vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No CNS-specific exposure data published. IDEAYA's GBM strategy depends on the IDE892 successor

(still phase 1, no GBM cohort yet) or the IDE397 + anvetmetostat combination (NCT05975073, completed) — the GBM relevance of IDE397 is the mechanism class, not the molecule. Logged because the broader MAT2A axis is part of the case's targetable feature mapping.

—/— (2024) — ClinicalTrials.gov record (Servier)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** S095035 (MAT2A inhibitor) monotherapy or + TNG462 (PRMT5 inhibitor)
- **Indication / model:** Advanced/metastatic MTAP-deleted solid tumors including IDH-wildtype GBM (explicit cohort) (2L+); MTAP homozygous deletion (archival tissue required)
- **Design:** Phase 1/2 open-label dose-escalation and expansion; S095035 monotherapy and S095035 + TNG462 combination arm
- **Effect size:** — — Phase 1: DLT, AEs, RP2D; Phase 2: ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024) — ClinicalTrials.gov record (Mirati / BMS)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MRTX1719 (PRMT5-MTA inhibitor)
- **Indication / model:** MTAP-deleted advanced solid tumors (mesothelioma, NSCLC, MPNST, pancreatic — no CNS) (2L+); MTAP homozygous deletion
- **Design:** Phase 1 multiple-cohort dose-escalation with 1b expansion
- **n:** 336
- **Effect size:** — — DLT, RP2D, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wen PY/Wen PY (2024) — ClinicalTrials.gov record (Dana-Farber)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Abemaciclib (CDK4/6 inhibitor, CNS-penetrant)
- **Indication / model:** Recurrent GBM or IDH-mutant WHO grade 4 astrocytoma at first relapse, CDKN2A/B-deleted or CDK4/6-amplified with intact RB (2L); CDKN2A/B (or C) homozygous deletion OR CDK4/6 high-level amplification AND wild-type RB
- **Design:** Phase 0/2 with paired surgical PK cohort
- **n:** 42
- **Effect size:** PFS6 approximately 9.7% in the non-surgical arm — Intratumoral abemaciclib concentration and PFS6
- **Variance:** 95% CI 2.5–22.9
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024) — ClinicalTrials.gov record (Penn)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CART-EGFRvIII + anti-IL13Ra2 (sequential/combined CAR product from O'Rourke lab)
- **Indication / model:** Recurrent EGFRvIII+ glioblastoma (2L+); EGFRvIII positive
- **Design:** Phase 1 dose-escalation of dual-target intrathecal CAR T
- **n:** 18
- **Effect size:** — — Safety, CRS/ICANS rate, MTD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024) — ClinicalTrials.gov record (NRG / NCI)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Ipilimumab + nivolumab + RT
- **Indication / model:** Newly diagnosed MGMT-unmethylated glioblastoma (NRG-BN007) (1L); MGMT unmethylated
- **Design:** Randomized phase 2/3 (phase 2 reported)
- **n:** 178
- **Effect size:** PFS HR 1.47 (immunotherapy worse than SOC) — OS / PFS (phase 2 analysis HR 1.47 for PFS — futility)
- **Variance:** p=0.96 (one-sided p for superiority)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wen PY/Trippa L (2023) — Clinical Cancer Research

Reference: [PMID 37722087](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Abemaciclib (INSIGHt platform arm)
- **Indication / model:** Newly diagnosed MGMT-unmethylated glioblastoma (1L (adjuvant, post-RT)); Adults with newly diagnosed MGMT-unmethylated IDH-wildtype GBM, post-resection or biopsy, completed standard chemoradiation. Bayesian adaptive randomization across abemaciclib, neratinib, CC-115, and control TMZ arms.
- **Design:** Phase 2 Bayesian adaptive platform trial (INSIGHt)
- **n:** 73
- **Effect size:** 0.72 HR — PFS (primary); OS (secondary)
- **Variance:** 95% CI 0.49–1.06; p=0.046 (one-sided)
- **Toxicities:** diarrhea; neutropenia G $\geq$ 3; fatigue
- **Case fit:** partial
- **Notes:** Inaugural INSIGHt readout. Abemaciclib produced a PFS signal but no OS benefit in MGMT-unmethylated newly-diagnosed GBM. Biomarker-stratified analyses (CDKN2A loss, intact RB) were planned but the n is too small to power subgroup conclusions. Relevant to this case because the patient has CDKN2A loss — the prespecified responder subgroup — but only enrollable if MGMT returns unmethylated.

Lassman AB/Aldape K (2023) — Neuro-Oncology

Reference: [PMID 35973199](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Depatuxizumab mafodotin (depatux-m, ABT-414, EGFR-directed ADC)
- **Indication / model:** Newly diagnosed EGFR-amplified glioblastoma (1L); Adults with newly diagnosed EGFR-amplified GBM (centrally confirmed), KPS ≥ 70 , post-resection or biopsy. Both MGMT-methylated and unmethylated enrolled.
- **Design:** Randomized double-blind phase 3 (INTELLANCE-1)
- **n:** 691
- **Effect size:** 1.02 HR — OS
- **Variance:** 95% CI 0.85–1.23; $p=0.63$
- **Toxicities:** corneal epitheliopathy / keratopathy; corneal epitheliopathy $G \geq 3$; blurred vision; treatment discontinuation due to ocular toxicity
- **Case fit:** weak
- **Notes:** INTELLANCE-1 negative phase 3 — depatux-m did not improve OS even in EGFR-amplified newly diagnosed GBM. Critical caveat that closes off this whole ADC class for first-line GBM despite the strong rationale (~40-50% of IDHwt GBM is EGFR-amplified). Ocular toxicity from the auristatin payload was severe. Decision-relevant context: the patient's EGFR amplification status is unknown but even if positive, the depatux-m era is over; the EGFR-directed energy is now in CAR-T (E-SYNC, Penn dual-target, MGH TEAM).

Bouffet E/Hargrave DR (2023) — N Engl J Med

Reference: [PMID 37733309](#) · **Type:** clinical · **Inclusion:** excluded — Pediatric low-grade glioma population (median age 9.5) — biology and clinical setting differ materially from adult IDHwt GBM. Adult HGG evidence in this patient flows through the ROAR HGG cohort (PMID 34838156).

- **Intervention:** Dabrafenib + trametinib in pediatric low-grade glioma
- **Notes:** Excluded: Pediatric low-grade glioma population (median age 9.5) — biology and clinical setting differ materially from adult IDHwt GBM. Adult HGG evidence in this patient flows through the ROAR HGG cohort (PMID 34838156). — Excluded from the synthesis fed to the board, but logged so the reviewer can see we considered the broader BRAF + MEK glioma evidence base.

Hargrave DR/Bouffet E (2023) — Journal of Clinical Oncology

Reference: [PMID 37643378](#) · **Type:** clinical · **Inclusion:** excluded — Pediatric high-grade glioma ($n=41$, median age 13). Biology and prior-therapy mix differ from adult IDHwt GBM. ORR 56% in this pediatric cohort is higher than the adult ROAR HGG signal — suggests biology-of-host matters — but adult evidence in this case rests on Wen 2022 (ROAR HGG).

- **Intervention:** Dabrafenib + trametinib in pediatric BRAF V600 HGG
- **Notes:** Excluded: Pediatric high-grade glioma ($n=41$, median age 13). Biology and prior-therapy mix differ from adult IDHwt GBM. ORR 56% in this pediatric cohort is higher than the adult ROAR HGG signal — suggests biology-of-host matters — but adult evidence in this case rests on Wen 2022 (ROAR HGG). — Pediatric companion to ROAR HGG; the higher pediatric ORR (56%) is informative for biology context but not directly applicable to this 66-year-old patient.

Omuro A/Lim M (2023) — Neuro-Oncology

Reference: [PMID 35419607](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Nivolumab + RT (CheckMate-498, MGMT unmethylated)

- **Indication / model:** Newly diagnosed MGMT-unmethylated glioblastoma (1L); Adults age 18+ with newly diagnosed MGMT-unmethylated GBM, KPS ≥ 70 , post-resection or biopsy. 560 randomized.
- **Design:** Randomized open-label phase 3
- **n:** 560
- **Effect size:** 1.31 HR — OS
- **Variance:** 95% CI 1.09–1.58; $p=0.0037$
- **Toxicities:** any immune-related AE; pneumonitis $G \geq 3$; thyroiditis; colitis $G \geq 3$; hepatitis $G \geq 3$
- **Case fit:** partial
- **Notes:** Negative phase 3 with HR favoring SOC. Substituting nivolumab for TMZ in MGMT-unmethylated newly diagnosed GBM made survival WORSE. Forecloses PD-1 monotherapy as a TMZ-sparing strategy in the unmethylated stratum.

Liau LM/Bosch ML (2023) — JAMA Oncology

Reference: [PMID 36394838](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** DCVax-L autologous dendritic-cell vaccine (Liau phase 3 ECC)
- **Indication / model:** Newly diagnosed and recurrent glioblastoma (1L (newly diagnosed) and 2L (recurrent)); 331 patients enrolled in the original phase 3 (232 DCVax-L + standard care; 99 randomized to standard care, with crossover allowed). External-control comparison used contemporaneous matched patients from 1,366 control-arm patients in other GBM trials.
- **Design:** Phase 3 prospective externally-controlled cohort trial (originally randomized with crossover)
- **n:** 331
- **Effect size:** 0.8 HR — OS
- **Variance:** 95% CI 0.7–0.89; $p < 0.001$
- **Toxicities:** injection-site reaction; flu-like symptoms; any TRAE $G \geq 3$
- **Case fit:** partial
- **Notes:** Methodology is heavily debated. The pivotal phase 3 was originally randomized with crossover; the published 2023 analysis substituted external matched controls instead of the originally-randomized comparator after protocol modifications. ROBINS-I serious risk-of-bias reflects this. The published OS signals (HR 0.80 newly diagnosed; HR 0.58 recurrent) are real but the magnitude is contested because the control choice was retrospective. Decision-relevant for this case because the user explicitly asked for compassionate-use options and DCVax-L has an expanded-access protocol (NCT02146066).

Ahluwalia MS/Fenstermaker RA (2023) — Journal of Clinical Oncology

Reference: [PMID 36473145](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** SurVaxM survivin peptide vaccine + adjuvant TMZ
- **Indication / model:** Newly diagnosed glioblastoma (1L (adjuvant)); 63 adults with newly diagnosed GBM who completed Stupp RT+TMZ without progression; KPS ≥ 70 ; survivin expression confirmed by IHC.
- **Design:** Open-label single-arm phase 2a multicenter
- **n:** 63
- **Effect size:** 95.2 % (PFS6) — PFS6 (primary), OS
- **Toxicities:** injection-site reaction; fatigue; any SAE attributable to SurVaxM $G \geq 3$ ($n=63$, 0%)
- **Case fit:** partial
- **Notes:** Phase 2a single-arm — the OS signal (mOS 25.9 mo) is encouraging vs the ~16-month historical control but single-arm with non-contemporaneous comparison. The pivotal randomized phase 2b (SURVIVE,

NCT05163080) is the trial that will determine whether this becomes real evidence. Eligible to enroll currently if SURVIVE has open slots; the SurVaxM expanded-access conversation is also available via MimiVax.

Ling AL/Chiocca EA (2023) — Nature

Reference: [PMID 37853118](#) · Type: clinical · Inclusion: included

- **Intervention:** CAN-3110 (rQNestin34.5v.2 oncolytic HSV-1)
- **Indication / model:** Recurrent high-grade glioma (mostly recurrent GBM) (2L+); 41 evaluable patients with recurrent HGG, KPS ≥ 70 , accessible enhancing tumor for stereotactic injection. Pre-existing HSV-1 serology was the key effect-modifier.
- **Design:** Phase 1 dose-escalation, single-center (Dana-Farber)
- **n:** 41
- **Effect size:** 11.6 months — Safety, OS, immune-activation biomarker
- **Variance:** 95% CI 7.8–14.9
- **Toxicities:** any DLT $G \geq 3$ (n=41, 0%); perilesional edema; headache; seizure; intratumoral hemorrhage
- **Case fit:** weak
- **Notes:** FDA Fast Track in 2024 for recurrent HGG. The 14.2-mo mOS in HSV-1-seropositive patients is the strongest signal — translates to $\sim 2\times$ the historical recurrent GBM benchmark. Single-dose, no maintenance. Recurrent-disease only; preposition for second-line conversation. The intratumoral hemorrhage history is a relative caution but not absolute — perilesional inflammation from viral lysis combined with prior bleed warrants explicit screening team conversation.

Sonabend AM/Stupp R (2023) — Lancet Oncology

Reference: [PMID 37087102](#) · Type: clinical · Inclusion: included

- **Intervention:** SonoCloud-9 microbubble-enhanced focused ultrasound BBB opening + nab-paclitaxel
- **Indication / model:** Recurrent glioblastoma — peri-tumoral BBB disruption to enable systemic chemotherapy delivery (2L+); 17 adults with recurrent GBM, KPS ≥ 60 , surgically implanted with the SonoCloud-9 skull device at time of resection. Single-center Northwestern.
- **Design:** Phase 1 dose-escalation, single-center
- **n:** 17
- **Effect size:** 4-6x increased peri-tumoral drug concentration vs unsonicated tissue; up to 6 cycles tolerated — Safety, feasibility, intracerebral nab-paclitaxel concentration
- **Toxicities:** implant-related complication $G \geq 3$ (n=17, 0%); FUS-related transient edema; nab-paclitaxel related peripheral neuropathy $G \geq 3$
- **Case fit:** weak
- **Notes:** First-in-human skull-implantable FUS device for repeated peri-tumoral BBB opening in GBM. The 4-6x drug-delivery enhancement is the proof-of-mechanism signal; the survival data is not yet mature. Decision-relevant for this case as the delivery-enhancement principle that could rescue any antibody-class or large-molecule CNS therapy (including pembrolizumab + FUS in the Leddon NCT07385846 dMMR trial).

Engstrom LD/Olson P (2023) — Cancer Discovery

Reference: [PMID 37552839](#) · Type: preclinical · Inclusion: included

- **Intervention:** MRTX1719 / navlimetostat — MTA-cooperative PRMT5 inhibitor

- **Indication / model:** MTAP-null vs MTAP-WT HCT116 isogenic pair; MTAP-deleted CDX panel (lung, pancreatic, bladder, mesothelioma); patient-derived xenografts; CSF/brain PK in rodent
- **Design:** MRTX1719 binds the PRMT5-MTA ternary complex preferentially over PRMT5-SAM, locking the enzyme in a non-productive state only when MTA is high (i.e. only in MTAP-null tumors). This is the mechanistic basis of the MTA-cooperative class — selectivity comes from substrate context, not enzyme isoform.
- **n:** Multiple CDX/PDX models (n=5-10 mice/arm typical); biochemical and cellular assays in replicate
- **Effect size:** strong — MRTX1719 had >70-fold selectivity for MTAP-null vs MTAP-WT HCT116 viability and produced dose-dependent SDMA reduction and tumor regression across MTAP-null xenografts. The paper is the chemical-biology proof of concept that an MTA-cooperative inhibitor avoids the hematologic toxicity that sank first-generation PRMT5 inhibitors.
- **Variance:** vs MTAP-WT isogenic and vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No GBM-specific in-vivo arm — the included tumor types are peripheral. MRTX1719 was lifecycle-managed into navlimetostat (BMS-986504) when BMS acquired Mirati; same molecule, current development under the MountainTAP program.

Ling AL/Chiocca EA (2023) — Nature

Reference:PMID 37853118 · **Type:** preclinical · **Inclusion:** excluded — Already on the clinical-evidence ledger (gbm-mtap-c4bq-can3110-ling-2023-nature). It is the phase 1 readout with immune-correlative tissue work, not a preclinical paper. The dedicated CAN-3110 preclinical toxicology paper (Chouljenko 2020 Mol Ther Methods Clin Dev) is the right preclinical row.

- **Intervention:** CAN-3110 (rQNestin34.5v.2) — clinical immunoactivation paper
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Already on the clinical-evidence ledger (gbm-mtap-c4bq-can3110-ling-2023-nature). It is the phase 1 readout with immune-correlative tissue work, not a preclinical paper. The dedicated CAN-3110 preclinical toxicology paper (Chouljenko 2020 Mol Ther Methods Clin Dev) is the right preclinical row.

Yang Y/Gromeier M (2023) — Neuro-Oncology

Reference:PMID 36864784 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PVSRIPO drives diffuse microglia activation across the CNS (immune-priming mechanism)
- **Indication / model:** Mouse intracerebral injection of PVSRIPO; immunohistochemistry and scRNA-seq of microglia and infiltrating myeloid cells
- **Design:** PVSRIPO triggers diffuse, sustained microglial activation extending well beyond the injection site through the CNS. The activation is type-I-interferon-dominant and persists for weeks, providing a prolonged immunostimulatory window in which adaptive antitumor responses can develop.
- **n:** Mouse cohorts n=6-10/group; multiple time points post-injection
- **Effect size:** strong — A single intratumoral PVSRIPO injection drove diffuse microglia activation through the CNS for weeks, with type-I-IFN gene signatures persisting beyond the local injection site. The finding explains the long-tail clinical responder pattern (Desjardins 2018 21% 24-month OS plateau) — the response is not from acute lysis alone but from delayed myeloid-driven adaptive immunity.
- **Variance:** vs Sham; UV-inactivated PVSRIPO; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial

- **Notes:** Mouse work — the human translation rests on CSF and tumor analysis from clinical samples, which the Brown/Gromeier group has begun publishing separately. The mechanism is the most coherent published explanation for why PVSRIPO has long-tail benefit despite a modest median-OS signal.

—/— (2023) — ClinicalTrials.gov record / AACR 2023 abstract (IDEAYA)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IDE397 (MAT2A inhibitor) monotherapy and combinations (e.g. + anvetmetostat — see NCT05975073)
- **Indication / model:** Advanced MTAP-deleted solid tumors (lung, bladder, others — no GBM cohort) (2L+); MTAP homozygous deletion
- **Design:** Phase 1 dose-escalation and expansion
- **Effect size:** — — DLT, RP2D, PK/PD
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wen PY/Subbiah V (2022) — Lancet Oncology

Reference: [PMID 34838156](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Dabrafenib + trametinib (ROAR HGG cohort)
- **Indication / model:** BRAF V600E-mutant high-grade glioma (HGG) — recurrent after standard therapy (2L+); Adults with BRAF V600E-mutant HGG (glioblastoma, anaplastic astrocytoma, anaplastic ganglioglioma, anaplastic pleomorphic xanthoastrocytoma), KPS ≥ 70 , prior RT and standard chemotherapy. 13 of 45 HGG patients had GBM.
- **Design:** Open-label single-arm phase 2 basket (ROAR)
- **n:** 45
- **Effect size:** 33 % — ORR (investigator-assessed RANO); OS
- **Variance:** 95% CI 20–49
- **Toxicities:** any grade ≥ 3 AE G ≥ 3 (n=58, 53%); fatigue G ≥ 3 (n=58, 9%); neutrophil count decreased G ≥ 3 (n=58, 9%); headache G ≥ 3 (n=58, 5%); neutropenia G ≥ 3 (n=58, 5%); pyrexia; rash / dermatitis
- **Case fit:** weak
- **Notes:** Tumor-agnostic FDA approval rests on this basket (with the pediatric data and other histologies). Eligible only if BRAF V600E confirms — rare in adult IDHwt GBM (~1-2%) but on the comprehensive NGS panel for this patient. The 33% ORR in HGG is substantial for a recurrent-glioma drug; the durability (mDoR 13.6 mo) is the standout vs typical recurrent-GBM trials.

Doz F/Drilon A (2022) — Neuro-Oncology

Reference: [PMID 34850167](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Larotrectinib in TRK fusion-positive primary CNS tumors
- **Indication / model:** TRK fusion-positive primary CNS tumors (including HGG, LGG, glioneuronal, others) (2L+); 33 patients with TRK fusion-positive primary CNS tumors from SCOUT (pediatric, NCT02637687) and NAVIGATE (adult, NCT02576431) pooled. 19 pediatric, 14 adult. 8 HGG, 11 LGG, others.
- **Design:** Pooled phase 1/2 expansion (CNS subset)
- **n:** 33

- **Effect size:** 30 % — ORR (investigator-assessed RANO)
- **Variance:** 95% CI 16–49
- **Toxicities:** any grade ≥ 3 AE $G \geq 3$; increased AST/ALT $G \geq 3$; dizziness
- **Case fit:** weak
- **Notes:** Largest published cohort of TRK inhibitor in primary CNS tumors. ORR 30% across mixed glioma histologies. NTRK fusions are vanishingly rare in adult IDHwt GBM ($< 1\%$) but the conditional gate is testable on the RNA-fusion panel that should be ordered with the comprehensive NGS workup.

Lim M/Reardon DA (2022) — Neuro-Oncology

Reference: [PMID 35511454](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Nivolumab + Stupp (CheckMate-548, MGMT methylated)
- **Indication / model:** Newly diagnosed MGMT-methylated glioblastoma (1L); Adults with newly diagnosed MGMT-methylated GBM by MSP, KPS ≥ 70 . 716 randomized.
- **Design:** Randomized double-blind phase 3
- **n:** 716
- **Effect size:** 1.1 HR — OS
- **Variance:** 95% CI 0.91–1.32; $p=0.34$
- **Toxicities:** any immune-related AE; pneumonitis $G \geq 3$; thyroiditis; colitis $G \geq 3$; hepatitis $G \geq 3$
- **Case fit:** partial
- **Notes:** Negative phase 3 — together with CheckMate-498 forecloses generic PD-1 + Stupp in either MGMT subgroup. The case for ICIs in GBM now rests on biomarker-selected populations (TMB-H, MSI-H, dMMR) or novel combinations (e.g. neoadjuvant pembrolizumab, FUS-BBB delivery).

Kalev P/Marjon K (2021) — Cancer Cell

Reference: [PMID 33450196](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** AG-270 (S095033) — first-in-class MAT2A inhibitor
- **Indication / model:** Isogenic MTAP knockout cell pairs across multiple lineages; MTAP-deleted PDX and CDX panel; RNA-seq and splicing analysis
- **Design:** AG-270 is a substrate-noncompetitive allosteric MAT2A inhibitor that blocks SAM release from the active site. Lowering SAM in MTA-rich MTAP-null cells removes the methyl donor PRMT5 needs to compete with MTA, collapsing residual PRMT5 activity below the lethal threshold. Downstream: PRMT5-dependent intron retention and DNA-damage accumulation.
- **n:** Multiple PDX/CDX models; $n=8-10$ mice/arm in efficacy studies
- **Effect size:** strong — AG-270 reduced intracellular SAM with IC₅₀ ~50 nM and selectively suppressed proliferation of MTAP-deleted cells and PDX tumors. Splicing defects and DNA damage were the proposed cell-biology consequences. AG-270 became S095033; the clinical phase 1 (Nat Commun 2024) was the first MAT2A inhibitor to translate.
- **Variance:** vs MTAP-WT isogenic; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No GBM-specific arm. AG-270 had clinical hepatotoxicity and was reformulated; the IDE397 / S095035 / IDE892 successors carry the cleaner mechanism forward. The paper is the foundational MAT2A pharmacology work that anchors the entire downstream class.

Sachamitr P/Dirks PB (2021) — Nature Communications

Reference:PMID 33479220 · Type: preclinical · Inclusion: included

- **Intervention:** PRMT5 inhibition disrupts splicing and stemness in GBM stem cells
- **Indication / model:** 46 patient-derived GBM stem cell cultures (proneural, mesenchymal, classical); orthotopic mouse xenografts; transcriptome + splicing analysis
- **Design:** PRMT5 catalyzes symmetric dimethylation of arginine in splicing-factor proteins; inhibition broadly disrupts pre-mRNA splicing, especially of cell-cycle genes. GBM stem cells are particularly sensitive because they rely on PRMT5-dependent splicing for self-renewal.
- **n:** 46 GSC lines; n=8-10 mice/arm in vivo
- **Effect size:** strong — PRMT5 inhibition (both Type I and Type II) suppressed proliferation across the 46-GSC panel, with proneural lines most sensitive. Widespread splicing changes — particularly intron retention in cell-cycle regulators — were the dominant transcriptome signature. LLY-283 is brain-penetrant and extended survival in orthotopic GSC xenografts.
- **Variance:** vs Vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Does not require MTAP loss for sensitivity — the dependency was broader. That weakens the case that PRMT5 inhibition is selective in MTAP-null GBM, but the MTA-cooperative drugs (TNG908 / AMG 193 / BGB-58067) restore the MTAP-selective therapeutic window. This paper is the strongest published GBM-specific PRMT5-inhibition dataset and is the model for why the brain-penetrant MTA-cooperative class is the right shape of drug.

Marabelle A/Doi T (2020) — Lancet Oncology

Reference:PMID 32919526 · Type: clinical · Inclusion: included

- **Intervention:** Pembrolizumab (TMB-H tumor-agnostic, KEYNOTE-158)
- **Indication / model:** TMB-H (≥ 10 mut/Mb) advanced solid tumors, tumor-agnostic (2L+); 102 patients with TMB-H by FoundationOne CDx from 10 KEYNOTE-158 tumor-type cohorts. Glioma not specifically in the primary TMB-H cohort.
- **Design:** Single-arm phase 2 basket (biomarker analysis)
- **n:** 102
- **Effect size:** 29 % — ORR (RECIST 1.1, BICR)
- **Variance:** 95% CI 21–39
- **Toxicities:** any immune-related AE; pneumonitis ≥ 3 ; colitis ≥ 3 ; thyroiditis (any); hepatitis ≥ 3
- **Case fit:** cross_tumor_only
- **Notes:** Tumor-agnostic FDA label for pembrolizumab in TMB-H ≥ 10 mut/Mb. Primary GBM is rarely TMB-H at diagnosis; hypermutation is more often acquired post-TMZ. The label is most likely to become relevant at recurrence if TMZ-induced hypermutation develops.

Reardon DA/Lim M (2020) — JAMA Oncology

Reference:PMID 32437507 · Type: clinical · Inclusion: included

- **Intervention:** Nivolumab vs bevacizumab (CheckMate-143, recurrent)
- **Indication / model:** Recurrent glioblastoma after Stupp (2L); 369 adults with first-recurrence GBM, KPS ≥ 70 ,

post-Stupp.

- **Design:** Randomized open-label phase 3
- **n:** 369
- **Effect size:** 1.04 HR — OS
- **Variance:** 95% CI 0.83–1.3; p=0.76
- **Toxicities:** any TRAE G_≥3 (n=182, 18.1%); any TRAE G_≥3 (n=165, 15.2%); pneumonitis G_≥3; fatigue
- **Case fit:** partial
- **Notes:** First randomized phase 3 of an ICI in primary brain tumors. Negative — nivolumab no better than bevacizumab in recurrent GBM. The 7.8% nivolumab ORR is the activity ceiling for ICI monotherapy in unselected recurrent GBM.

Cloughesy TF/— (2020) — Neuro-Oncology / sponsor disclosure

Reference: — · **Type:** clinical · **Inclusion:** excluded — Toca 511 program discontinued after the negative TOCA 5 phase 3 readout (Sept 2019, n=403). Drug is not enrollable; including for context only in a note on the oncolytic axis. The early-phase signal (Cloughesy 2018 Cancer Discovery) of durable CR did not survive the phase 3 confirmatory trial.

- **Intervention:** Toca 511 (vocimagene amiretrorepvec) + Toca FC (extended-release 5-FC)
- **Notes:** Excluded: Toca 511 program discontinued after the negative TOCA 5 phase 3 readout (Sept 2019, n=403). Drug is not enrollable; including for context only in a note on the oncolytic axis. The early-phase signal (Cloughesy 2018 Cancer Discovery) of durable CR did not survive the phase 3 confirmatory trial. — Sponsor Tocagen discontinued the program. Logged as audit-trail context for why the oncolytic-virus axis is concentrated on CAN-3110 and PVSRIPO and not on the broader Toca platform.

Chiocca EA/Oglesbee M (2020) — Molecular Therapy — Methods & Clinical Development

Reference: [PMID 32373649](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CAN-3110 (rQNestin34.5v.2) preclinical toxicology and design rationale
- **Indication / model:** Murine intracranial glioma model; non-human primate biodistribution; nestin-promoter glioma vs normal-brain cell selectivity assays
- **Design:** rQNestin34.5v.2 places the neurovirulence gene ICP34.5 under control of the nestin promoter, which is selectively active in high-grade glioma and glioma stem cells. The virus replicates and lyses tumor in nestin-expressing cells, while normal CNS cells (which don't drive the nestin promoter) restrict ICP34.5 expression and avoid neurovirulence.
- **n:** Mouse cohorts n=8-10/arm; NHP biodistribution n=3
- **Effect size:** strong — rQNestin34.5v.2 was cytotoxic for all tested glioma cell lines (cell survival <20% of control) while sparing nestin-low normal CNS cells. Mouse and non-human primate biodistribution showed restricted replication in nestin-active tissue. The toxicology package supported the IND for the NCT03152318 phase 1.
- **Variance:** vs Sham injection; parental HSV-1 backbones; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Nestin expression is not glioma-specific (also expressed in neural stem cells and some adult progenitors) — the safety margin comes from differential promoter activity, not absolute selectivity. The preclinical toxicology package is the most thorough in the modern oncolytic-HSV space and is the backstop for the clinical Nature 2023 readout.

Akkari L/Joyce JA (2020) — Science Translational Medicine

Reference:PMID 32669424 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Dynamic macrophage repopulation after RT — CSF-1R as a resistance node
- **Indication / model:** PDGFB-driven glioma GEMM with fractionated cranial RT; CSF-1R blockade combined with RT
- **Design:** After RT, monocyte-derived macrophages repopulate the tumor and adopt a wound-healing / pro-tumorigenic program that drives recurrence. CSF-1R blockade during the post-RT window blocks this repopulation and prevents the resistance trajectory.
- **n:** n=8-12 mice/arm across multiple GEMM cohorts
- **Effect size:** strong — Adding CSF-1R blockade to RT in PDGFB GEMM glioma extended survival far beyond either modality alone. The post-RT macrophage influx was identified as a tractable resistance node — same biology as the immune-suppressive TME in untreated tumors but kinetically amplified after radiation injury.
- **Variance:** vs RT alone; CSF-1R alone; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** GEMM models — clinical translation of the CSF-1R + RT concept has not yet shown a definitive phase 2/3 signal. Mechanism is the part to carry forward: it explains why standard RT in GBM almost always leads to recurrence with an adaptive immune escape that the patient's IDH-WT GBM will face on its Stupp track.

Cao Y/Liu Y (2020) — Journal of Cellular and Molecular Medicine

Reference:PMID 32277580 · **Type:** preclinical · **Inclusion:** excluded — Paper was retracted post-publication for image-integrity concerns. The published mechanistic claim (TMZ-induced p-Rb reversed by abemaciclib) is plausible but the data backing the survival numbers is not reliable. Excluded from synthesis. The same combination logic is supported by Raub 2015 (orthotopic abemaciclib activity) and is being tested empirically in INSIGHT.

- **Intervention:** CDK4/6 + temozolomide in glioma — retracted paper
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Paper was retracted post-publication for image-integrity concerns. The published mechanistic claim (TMZ-induced p-Rb reversed by abemaciclib) is plausible but the data backing the survival numbers is not reliable. Excluded from synthesis. The same combination logic is supported by Raub 2015 (orthotopic abemaciclib activity) and is being tested empirically in INSIGHT.

Klemm F/Joyce JA (2020) — Cell

Reference:PMID 32470396 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Microenvironmental atlas of brain tumor myeloid compartment (IDH-WT bias)
- **Indication / model:** Human brain tumor resections (gliomas IDH-WT and IDH-mutant, brain metastases); flow cytometry, RNA-seq, mass cytometry across 119 patients
- **Design:** IDH-WT GBM is dominated by tissue-resident microglia and monocyte-derived macrophages that express immunosuppressive markers (TGFB1, CD163, MARCO) and lack the activation signatures seen in IDH-mutant tumors or brain metastases. T cells are sparse and exhausted; the dominant immune-cell pool is myeloid.
- **n:** 119 patients across glioma subtypes and brain metastasis

- **Effect size:** strong — IDH-WT GBM has a uniquely myeloid-dominated immune microenvironment with minimal T-cell infiltration — substantially more immunosuppressive than IDH-mutant glioma or brain metastases. The signature predicts why generic PD-1 blockade fails (no T cells to unleash) and why myeloid-targeted strategies (CSF-1R, TGF- β , CD47) and locoregional T-cell delivery (intraventricular CAR-T) are the mechanistically supported alternatives.
- **Variance:** vs Brain metastases and IDH-mutant glioma as biology comparators; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive immunology — no therapeutic intervention. The IDH-WT vs IDH-mutant contrast is the part most directly relevant to the patient (who is IDH-WT). Justifies the case-level skepticism about generic ICI and the case-level interest in neoadjuvant pembrolizumab (Cloughesy 2019) and oncolytic-virus immune priming (CAN-3110).

Herrlinger U/Glas M (2019) — Lancet

Reference: [PMID 31488360](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Lomustine + temozolomide + radiotherapy (CeTeG / NOA-09)
- **Indication / model:** Newly diagnosed MGMT-methylated glioblastoma (1L); Adults age 18-70, KPS ≥ 70 , newly diagnosed GBM with MGMT promoter methylated by MSP, post-resection or biopsy. 17 German university centers.
- **Design:** Randomized open-label phase 3
- **n:** 141
- **Effect size:** 0.6 HR — OS (per-protocol)
- **Variance:** 95% CI 0.35–1.03; $p=0.049$
- **Toxicities:** any grade 3-4 hematologic AE ≥ 3 (n=63, 41%); lymphopenia ≥ 3 (n=63, 27%); thrombocytopenia ≥ 3 (n=63, 16%); neutropenia ≥ 3 (n=63, 13%); any grade 3-4 hematologic AE ≥ 3 (n=66, 13%)
- **Case fit:** partial
- **Notes:** Eligible only if MGMT methylation testing returns methylated. Open-label and small sample size (n=141) are the caveats; the OS gain of 16.7 months is the largest published signal in newly diagnosed MGMT-methylated GBM. Cumulative CCNU myelosuppression is the practical limiter at age 66.

Goff SL/Rosenberg SA (2019) — Journal of Immunotherapy

Reference: [PMID 30418325](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** CART-EGFRvIII (NCI Clone 139 construct + lymphodepletion)
- **Indication / model:** Recurrent EGFRvIII-positive glioblastoma (2L+); 18 adults with EGFRvIII+ recurrent GBM, KPS ≥ 60 ; lymphodepletion pre-conditioning before CAR-T infusion. NCI Surgery Branch.
- **Design:** Phase 1/2 dose-escalation, single-center
- **n:** 18
- **Effect size:** 1 long-term survivor at 59 months; no objective RANO responses; 1 treatment-related death at high dose ($>1 \times 10^{10}$ CD3+ cells) — Safety, response, persistence
- **Toxicities:** treatment-related death G5 (1/18); CRS ≥ 3 (n=17, 0%); lymphodepletion-related cytopenias ≥ 3
- **Case fit:** weak
- **Notes:** NCI/Rosenberg-lab predecessor program to the Penn dual-target work. Lessons: (a) systemic CAR delivery has a strict dose ceiling for CNS targets, (b) lymphodepletion does not by itself solve the trafficking +

persistence problem, (c) single-antigen targeting permits antigen escape. The 1/18 long-term survivor is the same finding pattern as the O'Rourke STM 2017 paper — durable responses are possible but rare with first-generation single-antigen constructs.

Cloughesy TF/Prins RM (2019) — Nature Medicine

Reference:PMID 30742122 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Neoadjuvant pembrolizumab (recurrent GBM)
- **Indication / model:** Recurrent surgically-resectable glioblastoma (2L+); 35 adults with recurrent GBM amenable to repeat resection, KPS ≥ 70 . Randomized 16 to neoadjuvant + adjuvant pembrolizumab and 19 to adjuvant-only.
- **Design:** Randomized phase 2
- **n:** 35
- **Effect size:** 13.7 months — OS, intratumoral immune correlates
- **Variance:** $p=0.04$
- **Toxicities:** any TRAE $G \geq 3$; perioperative complication
- **Case fit:** weak
- **Notes:** Small randomized trial ($n=35$) — hypothesis-generating signal that neoadjuvant ICI timing matters more than adjuvant-only in GBM. The OS gain (mOS 13.7 vs 7.5 mo) and the intratumoral immune correlates are the most-cited PD-1 GBM signal. Not enrollable here (recurrent disease), but the principle is influencing trial design across the GBM ICI space.

Tien AC/Sanai N (2019) — Clinical Cancer Research

Reference:PMID 31285369 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Ribociclib CNS penetration in recurrent GBM (phase 0 PK/PD)
- **Indication / model:** Phase 0 trial in recurrent GBM with intraoperative tumor sampling; matched plasma and tumor PK; Rb phosphorylation as PD readout
- **Design:** Ribociclib is a CDK4/6 inhibitor with lower P-gp affinity than palbociclib but higher than abemaciclib. The phase 0 design measured whether intratumoral concentrations exceeded the in-vitro K_i — the only way to confirm that a CDK4/6 inhibitor is reaching its target in human GBM.
- **n:** $n=12$ patients (3 surgical + 9 expansion)
- **Effect size:** moderate — Ribociclib reached pharmacologically active concentrations in both enhancing and non-enhancing tumor regions and produced measurable inhibition of Rb phosphorylation. This is the first human PK/PD demonstration that a CDK4/6 inhibitor can engage its target in GBM tissue.
- **Variance:** vs Pre-treatment baseline (paired); translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Phase 0 — designed for PK/PD, not efficacy. The Rb phosphorylation suppression was target engagement, not tumor response. Bridges the rodent preclinical (Raub 2015) and the clinical-activity question (INSIGHt / Wen NCT02981940). Confirms that ribociclib joins abemaciclib as the CDK4/6 inhibitor pair with adequate CNS exposure; palbociclib falls off the list.

Choi BD/Maus MV (2019) — Nature Biotechnology

Reference:PMID 31332324 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CAR-T secreting bispecific T-cell engagers (TEAMs) — preclinical of the MGH construct
- **Indication / model:** NSG mouse intracranial U87-EGFRvIII / WT EGFR mixed xenograft; antigen-heterogeneous tumor model
- **Design:** CAR targets EGFRvIII directly through the CAR; the engineered T cells also secrete a CD3 x WT-EGFR bispecific antibody (the TEAM) that redirects endogenous T cells against neighboring EGFRvIII-negative, WT-EGFR-positive tumor cells. The TEAM solves the antigen-escape problem the O'Rourke 2017 trial exposed.
- **n:** n=8-10 mice/arm
- **Effect size:** strong — CAR-T-TEAM cells eradicated mixed antigen-heterogeneous intracranial tumors that single-target EGFRvIII CAR-T cells could not. The paper is the direct preclinical blueprint for the Choi 2024 NEJM CARv3-TEAM-E first-in-human readout — which showed dramatic radiographic responses on the same logic.
- **Variance:** vs EGFRvIII CAR-T without TEAM secretion; untransduced T cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Xenograft only, no syngeneic / immunocompetent arm. WT EGFR is not expressed in normal mouse brain in the model, so the on-target/off-tumor risk for human CNS (normal mouse brain has low WT-EGFR but human brain endothelium does) is not tested. Intraventricular delivery in the clinical trial mitigates systemic exposure but the design caveat is real.

Vilgelm AE/Richmond A (2019) — PNAS

Reference: [PMID 31439820](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRMT5 + CDK4/6 combination — preclinical synergy (cross-tumor with GBM implications)
- **Indication / model:** Melanoma cell lines and PDX (MTAP status mixed); CDK4/6 inhibitor (palbociclib) + PRMT5 inhibitor (GSK3326595) in vitro and in vivo; MDM4 splicing as molecular endpoint
- **Design:** CDK4/6 inhibition indirectly suppresses PRMT5 activity through cell-cycle-dependent regulation of PRMT5 cofactors; PRMT5 in turn regulates MDM4 pre-mRNA splicing, which controls p53 activity. The MDM4 axis links the two enzymes — dual inhibition converges on p53 reactivation and synthetic lethality.
- **n:** Multiple lines and PDX; n=8-10 mice/arm in vivo
- **Effect size:** strong — Combined palbociclib + GSK3326595 produced deeper and more durable regressions than either agent alone in melanoma models, and delayed emergence of CDK4/6i resistance. The MDM4-p53 axis was the proposed convergent node. Subsequent work in breast cancer (Wang 2024 Nat Commun PMID 38493145) extended the same logic to ER+/Rb-deficient tumors.
- **Variance:** vs Either agent alone; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Melanoma, not GBM. The biology — MDM4 splicing and p53 reactivation — is general but the GBM-specific test has not been done in published preclinical work. Relevant to the case because TNG462 has reported preclinical combination signals with CDK4/6 inhibitors in MTAP-null preclinical models (AACR 2025 abstract 2996), and a PRMT5i + CDK4/6i combination in CDKN2A/MTAP co-deleted GBM is the most mechanistically attractive doublet on this patient's chart.

Taal W/Taphoorn MJB (2018) — Journal of Neuro-Oncology

Reference: [PMID 30151703](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Palbociclib (recurrent RB1-positive GBM)
- **Indication / model:** Recurrent glioblastoma, RB1-proficient by IHC (2L+); Adults with recurrent GBM, KPS ≥ 60 , RB1 protein expression confirmed by IHC, ≤ 3 prior relapses. Arm 1 = palbociclib pre-op then adjuvant; Arm 2 = palbociclib without resection.
- **Design:** Open-label single-arm phase 2 (two-arm design with a paired-surgical cohort)
- **n:** 22
- **Effect size:** 1.2 months — PFS, OS
- **Toxicities:** neutropenia ≥ 3 ; thrombocytopenia ≥ 3 ; fatigue
- **Case fit:** weak
- **Notes:** Negative trial — closed early for futility. The contrast with the abemaciclib recurrent GBM trial (Wen, NCT02981940) is informative: palbociclib has poor CNS penetration (Pgp substrate) and the trial did not require CDKN2A loss; abemaciclib has substantially better CNS penetration and the trial was biomarker-selected. The right lesson is that CDK4/6 inhibitor selection AND biomarker selection both matter — palbociclib failure does not refute the CDK4/6 pathway in CDKN2A-loss GBM.

Drilon A/Hyman DM (2018) — N Engl J Med

Reference: [PMID 29466156](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Larotrectinib (tumor-agnostic NTRK basket)
- **Indication / model:** TRK fusion-positive advanced solid tumors (tumor-agnostic) (2L+); 55 patients (children + adults) with TRK fusion-positive tumors pooled from three early-phase trials. 17 tumor types represented; primary CNS not in this initial pool.
- **Design:** Pooled phase 1/2 three-trial basket
- **n:** 55
- **Effect size:** 75 % — ORR (investigator + central review)
- **Variance:** 95% CI 61–85
- **Toxicities:** any grade ≥ 3 AE ≥ 3 ; anemia ≥ 3 ; increased AST ≥ 3 (n=55, 3%); increased ALT ≥ 3 (n=55, 3%); dizziness (n=55, 22%)
- **Case fit:** cross_tumor_only
- **Notes:** Foundational tumor-agnostic basket supporting FDA approval. Adult IDHwt GBM was not in the initial pool but CNS responses were demonstrated in subsequent expansion (Doz 2022).

Desjardins A/Bigner DD (2018) — N Engl J Med

Reference: [PMID 29943666](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Lerapolturev (PVSRIPO, oncolytic recombinant poliovirus chimera)
- **Indication / model:** Recurrent WHO grade IV glioblastoma (2L+); 61 adults with recurrent supratentorial GBM, KPS ≥ 70 , single contrast-enhancing lesion 1-5 cm amenable to CED catheter placement. Single-center Duke.
- **Design:** Phase 1 dose-escalation, single-center
- **n:** 61
- **Effect size:** 12.5 months — Safety, OS
- **Variance:** 95% CI 9.9–15.2
- **Toxicities:** peritumoral inflammation / edema; peritumoral inflammation ≥ 3 (n=61, 19%); hemiparesis ≥ 3 ; seizure; PVSRIPO-related neuropoliomyelitis (n=61, 0%)
- **Case fit:** weak
- **Notes:** The 21% OS plateau at 24 and 36 months is the standout finding — substantially above historical

recurrent GBM controls (~14%). Phase 2 readout (Desjardins 2025, PMID 40313264) confirmed mOS ~7-12 mo across cohorts and the long-tail plateau. Recurrent disease only; CED delivery requires accessible single lesion. Pre-position for second-line conversation.

Banasavadi-Siddegowda YK/Kaur B (2018) — Neuro-Oncology

Reference:PMID 29106602 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRMT5 as a druggable target in GBM (Banasavadi-Siddegowda)
- **Indication / model:** GBM neurospheres (patient-derived); orthotopic mouse xenograft
- **Design:** PRMT5 is overexpressed in GBM and supports tumor cell survival, partly through E2F1 and p53 pathway methylation. Knockdown or inhibition reduces self-renewal and induces apoptosis in neurospheres.
- **n:** Multiple neurosphere lines; n=8 mice/arm in vivo
- **Effect size:** moderate — PRMT5 knockdown or pharmacologic inhibition (EPZ015666) suppressed GBM neurosphere growth in vitro and prolonged survival in orthotopic xenografts. The paper helped establish PRMT5 as a GBM-relevant target — preceded the MTAP-deletion synthetic-lethality wave and motivated the GBM-focused successor drugs.
- **Variance:** vs Vehicle or shRNA control; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Uses first-generation PRMT5 inhibitor (EPZ015666) without MTAP selectivity — the off-target hematologic toxicity that capped EPZ015666's clinical dose is the reason MTA-cooperative successors became necessary. Mechanism evidence holds; drug specifically does not translate.

Lang FF/Fueyo J (2018) — Journal of Clinical Oncology

Reference:PMID 29432077 · **Type:** preclinical · **Inclusion:** excluded — Clinical phase 1 paper with mechanism backdrop, not a preclinical paper. DNX-2401 is not on the patient's intervention list (not in trials.jsonl or clinical_evidence.jsonl) — including a preclinical row for it would be off-scope. Logged as audit trail; the broader oncolytic-virus class is covered by CAN-3110 and PVSRIPO preclinical rows.

- **Intervention:** DNX-2401 / Delta-24-RGD oncolytic adenovirus — clinical with mechanism context
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Clinical phase 1 paper with mechanism backdrop, not a preclinical paper. DNX-2401 is not on the patient's intervention list (not in trials.jsonl or clinical_evidence.jsonl) — including a preclinical row for it would be off-scope. Logged as audit trail; the broader oncolytic-virus class is covered by CAN-3110 and PVSRIPO preclinical rows.

Stupp R/Ram Z (2017) — JAMA

Reference:PMID 29392280 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tumor-treating fields (Optune) + adjuvant temozolomide
- **Indication / model:** Newly diagnosed glioblastoma post-RT+TMZ; adjuvant maintenance (maintenance); Age 18+, completed RT+TMZ without progression, KPS $\geq$ 70, both MGMT methylated and unmethylated enrolled.
- **Design:** Randomized open-label phase 3 (EF-14)
- **n:** 695
- **Effect size:** 0.63 HR — OS

- **Variance:** 95% CI 0.53–0.76; $p < 0.001$
- **Toxicities:** scalp dermatitis / irritation under arrays (n=466, 52%); scalp dermatitis $G \geq 3$ (n=466, 2%); thrombocytopenia $G \geq 3$ (n=466, 4%); fatigue
- **Case fit:** strong
- **Notes:** EF-14 pivotal trial supporting FDA-approved Optune indication. Open-label is the main bias risk. The OS benefit holds across MGMT subgroups but is larger in methylated. Practical commitment is the ≥ 18 -hr/day wear-time.

Perry JR/Laperriere N (2017) — N Engl J Med

Reference: [PMID 28296618](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Hypofractionated radiotherapy + temozolomide (Perry / CCTG CE.6)
- **Indication / model:** Newly diagnosed glioblastoma in adults age ≥ 65 (1L); Age ≥ 65 , newly diagnosed GBM, ECOG 0-2 (KPS ≥ 50). MGMT methylation tested in 354 of 562 patients with available tissue.
- **Design:** Randomized phase 3 (CCTG CE.6 / EORTC 26062-22061)
- **n:** 562
- **Effect size:** 0.67 HR — OS
- **Variance:** 95% CI 0.56–0.8; $p < 0.001$
- **Toxicities:** lymphopenia $G \geq 3$ (n=281, 27%); thrombocytopenia $G \geq 3$ (n=281, 10%); neutropenia $G \geq 3$ (n=281, 5%); nausea $G \geq 3$ (n=281, 5%); vomiting $G \geq 3$ (n=281, 5%); constipation $G \geq 3$ (n=281, 3%)
- **Case fit:** partial
- **Notes:** Patient is 66 — meets inclusion. At ECOG 1 the patient could tolerate full-course Stupp; this Perry regimen is the elderly-GBM alternative when fitness or access argue for a shorter 3-week RT course. MGMT-methylated subgroup derives the largest benefit (HR 0.53).

O'Rourke DM/Maus MV (2017) — Science Translational Medicine

Reference: [PMID 28724573](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** CART-EGFRvIII (first-generation Penn construct)
- **Indication / model:** Recurrent EGFRvIII-positive glioblastoma (2L+); 10 adults with recurrent EGFRvIII+ GBM by RT-PCR; 7 had subsequent surgical resection enabling paired tumor and T-cell analysis.
- **Design:** Phase 1 single-arm pilot, single-center (Penn)
- **n:** 10
- **Effect size:** Demonstrable CAR-T trafficking to tumor in 7/7 resected; selective EGFRvIII antigen loss in tumor post-infusion; 1/10 stable disease at 18 mo — — Safety, CAR-T trafficking, EGFRvIII antigen loss
- **Toxicities:** CRS $G \geq 3$ (n=10, 0%); neurologic AE (cerebral edema, seizure) $G \geq 3$; any TRAE $G \geq 3$
- **Case fit:** weak
- **Notes:** Foundational paper for EGFRvIII CAR-T in GBM — demonstrated feasibility and antigen-specific tumor pressure, but also the adaptive resistance via EGFRvIII downregulation and tumor-microenvironment immunosuppression. The 1/10 durable SD signal and the resistance findings together motivated the dual-target Penn successor program (NCT05802693 and NCT07209241). Per-term AE table is not in the main paper text — sourced from main-text descriptions; user-supplied AE table or supplementary file would be needed to backfill specific events. The Penn lab safety profile across the EGFRvIII CAR program is the broader context.

Weller M/Reardon DA (2017) — Lancet Oncology

Reference:PMID 28844499 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Rindopepimut (EGFRvIII peptide vaccine, ACT IV)
- **Indication / model:** Newly diagnosed EGFRvIII-expressing glioblastoma, minimal residual disease post-RT+TMZ (1L (adjuvant)); 745 adults with EGFRvIII-positive newly diagnosed GBM, post-resection ($\geq$ GTR or minimal residual disease) and completed RT + TMZ without progression. 165 sites in 22 countries.
- **Design:** Randomized double-blind international phase 3
- **n:** 745
- **Effect size:** 1.01 HR — OS
- **Variance:** 95% CI 0.79–1.3; $p=0.93$
- **Toxicities:** injection-site reaction; any TRAE $G\geq 3$
- **Case fit:** weak
- **Notes:** Negative phase 3 — rindopepimut did not improve OS in EGFRvIII+ newly diagnosed GBM. Critical caveat for any EGFRvIII-targeted vaccine strategy. The leading hypothesis for failure is antigen escape (EGFRvIII heterogeneous within tumor, lost on relapse) — the same mechanism that limited the first-generation Penn EGFRvIII CAR-T. The lesson influenced subsequent dual-target (EGFR + IL13Ra2) and synNotch designs in this case's E-SYNC and CART-EGFR-IL13Ra2 evidence rows.

Wick W/Gorlia T (2017) — N Engl J Med

Reference:PMID 29091557 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Bevacizumab + lomustine (EORTC 26101, recurrent GBM)
- **Indication / model:** Recurrent glioblastoma after Stupp (2L); 437 adults with recurrent GBM, WHO PS 0-2, post-Stupp.
- **Design:** Randomized phase 3
- **n:** 437
- **Effect size:** 0.95 HR — OS
- **Variance:** 95% CI 0.74–1.21; $p=0.65$
- **Toxicities:** any grade ≥ 3 AE $G\geq 3$ ($n=288$, 64%); any grade ≥ 3 AE $G\geq 3$ ($n=149$, 38%); neutropenia $G\geq 3$; thrombocytopenia $G\geq 3$; hypertension $G\geq 3$; intracranial hemorrhage $G\geq 3$; venous thromboembolism $G\geq 3$; treatment-related death G5 (4/288)
- **Case fit:** partial
- **Notes:** PFS gain but no OS benefit — the published statement of recurrent-GBM bevacizumab activity. Together with BRAIN it defines the recurrence bevacizumab option set. Same intratumoral-hemorrhage caveat applies as on the Friedman 2009 row.

Li M/Purow B (2017) — Oncotarget

Reference:PMID 28903422 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CDK4/6 inhibition is subtype-selective in GBM stem cells
- **Indication / model:** Patient-derived glioblastoma stem cell (GSC) cultures across proneural / mesenchymal / classical subtypes; orthotopic mouse xenograft
- **Design:** Proneural GSCs express higher CDK6 and have intact Rb1, putting them in the canonical CDK4/6-inhibitor-sensitive zone. CDK4-overexpressing / Rb1-null lines are resistant. Subtype matters: same drug, different response by transcriptional state.
- **n:** Multiple GSC lines ($n=12+$); $n=8-10$ mice/arm orthotopic

- **Effect size:** moderate — Palbociclib preferentially inhibited proneural GSCs in vitro and extended survival in a proneural-line orthotopic xenograft. Importantly, treatment also pushed proneural GSCs toward a mesenchymal expression program — a hint that CDK4/6 inhibition may select for or induce the more aggressive subtype on prolonged therapy.
- **Variance:** vs Vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Uses palbociclib (the BBB-poor compound) — the in-vivo orthotopic signal is more impressive than the Parrish 2015 work would predict, possibly because of the proneural-line bias. The proneural-to-mesenchymal shift on therapy is the part to carry forward; it predicts that monotherapy CDK4/6 inhibition may have a built-in resistance trajectory.

O'Rourke DM/Maus MV (2017) — Science Translational Medicine

Reference: [PMID 28724573](#) · **Type:** preclinical · **Inclusion:** excluded — Already captured as a clinical-evidence row (gbm-mtap-c4bq-egfrviii-orourke-2017-stm). The intratumoral correlative biology (PD-L1, IDO, Treg upregulation post-CAR-T) is genuinely preclinical in feel but the paper is a clinical phase 1 with paired-tissue analysis, so it belongs to the clinical dossier.

- **Intervention:** O'Rourke 2017 STM (clinical paper with preclinical adaptive-resistance correlates)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Already captured as a clinical-evidence row (gbm-mtap-c4bq-egfrviii-orourke-2017-stm). The intratumoral correlative biology (PD-L1, IDO, Treg upregulation post-CAR-T) is genuinely preclinical in feel but the paper is a clinical phase 1 with paired-tissue analysis, so it belongs to the clinical dossier.

Brown MC/Nair SK (2017) — Science Translational Medicine

Reference: [PMID 28931654](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PVSRIPO mechanism — CD155-targeted oncolysis and myeloid immunoactivation
- **Indication / model:** Human dendritic cell and macrophage cultures; CD155-expressing tumor lines (glioma, melanoma); ex-vivo tumor explants
- **Design:** PVSRIPO is a recombinant poliovirus chimera with the internal ribosomal entry site swapped from rhinovirus type 2, ablating neuropathogenicity. CD155 (the poliovirus receptor) is ectopically expressed on glioma cells and constitutively expressed on glioma-associated macrophages and microglia. Sublethal infection of CD155+ antigen-presenting cells produces sustained type-I-interferon-dominant activation.
- **n:** Multiple human DC and macrophage donors; cell lines in replicate
- **Effect size:** strong — Sublethal PVSRIPO infection of CD155+ dendritic cells and macrophages produced sustained type-I-IFN-dominant activation and primed antigen-specific CTL responses against tumor neoantigens. The dual mechanism (direct lysis of tumor cells + immune-priming of antigen-presenting cells) is the proposed reason for the long-tail responder pattern seen clinically in the Desjardins 2018 NEJM phase 1.
- **Variance:** vs Wild-type poliovirus; CD155-negative cells; non-replicating virus; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Mechanism paper, mostly in-vitro. The in-vivo confirmation of myeloid activation came later (Mosaheb 2020 Neuro-Oncology). The CD155 expression data is the part to read carefully — broad CD155 expression on tumor and myeloid cells is exactly what you want, but it is not glioma-specific.

Wick W/Gorlia T (2017) — N Engl J Med

Reference:PMID 32720870 · Type: trial · Inclusion: included

- **Intervention:** Lomustine + bevacizumab
- **Indication / model:** Recurrent glioblastoma after Stupp (2L); None required
- **Design:** Randomized phase 3 (EORTC 26101)
- **n:** 437
- **Effect size:** median OS 9.1 vs 8.6 mo; HR 0.95 (no OS benefit) — OS
- **Variance:** 95% CI 0.74–1.21; p=0.65
- **Case fit:** weak
- **Notes:** (toxicity table not extracted)

Kryukov GV/Garraway LA (2016) — Science

Reference:PMID 26912360 · Type: preclinical · Inclusion: included

- **Intervention:** PRMT5 inhibition (class evidence — MTAP-loss synthetic lethality)
- **Indication / model:** Pan-cancer cell-line panel (Project Achilles + Project Drive); CCLE pharmacogenomic correlations; isogenic MTAP knockout pairs
- **Design:** MTAP loss causes accumulation of methylthioadenosine (MTA), the MTAP substrate, which binds PRMT5 and partially blocks its SAM-dependent methyltransferase activity. Because PRMT5 is essential, the partial inhibition leaves MTAP-null cells closer to the lethal threshold and selectively sensitive to additional PRMT5 knockdown.
- **n:** 390+ cell lines screened across two shRNA datasets; isogenic validation in 4 lineages
- **Effect size:** strong — PRMT5 dependency in 9p21-deleted cell lines was the strongest synthetic-lethal hit from the pan-cancer shRNA screen, with MTA accumulation as the proposed metabolic explanation. The finding nominated PRMT5 as the lead target for MTAP-deleted tumors and set the rationale for the entire MTA-cooperative PRMT5 inhibitor program.
- **Variance:** vs MTAP-wildtype isogenic and non-isogenic comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Genetic knockdown, not small-molecule pharmacology. No GBM-specific cohort in the screen; GBM lines are represented but mixed into the pan-cancer dependency map. This is the foundational paper for the entire class — every clinical PRMT5 program traces back here.

Mavrakis KJ/Sellers WR (2016) — Science

Reference:PMID 26912361 · Type: preclinical · Inclusion: included

- **Intervention:** PRMT5 inhibition (Mavrakis companion paper — independent confirmation)
- **Indication / model:** Pan-cancer shRNA screen across ~390 cell lines (Project Drive); validation in MTAP-deleted vs MTAP-WT isogenic pairs (HCT116, A549, others)
- **Design:** Independent shRNA screen converged on MAT2A and PRMT5 as essential in MTAP-deleted cells. MAT2A produces SAM, the methyl donor; lowering SAM in MTA-rich MTAP-null cells reduces residual PRMT5 activity past the lethal threshold.
- **n:** ~390 cell lines screened; ~10 isogenic pairs for validation
- **Effect size:** strong — PRMT5 and MAT2A both score as selective dependencies in MTAP-deleted cells. RIOK1

emerged as a downstream PRMT5-regulated effector. Two nodes (PRMT5, MAT2A) and a downstream sensor (RIOK1) were nominated in the same paper, broadening the therapeutic axis.

- **Variance:** vs MTAP-WT isogenic lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Companion paper to Kryukov 2016 — published same day, same journal. Together with Marjon 2016 these three papers are the foundational triplet. Pharmacology came later.

Marjon K/Marks KM (2016) — Cell Reports

Reference:PMID 27068473 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MAT2A and PRMT5 axis (Marjon 2016 — third foundational paper)
- **Indication / model:** shRNA screen + isogenic MTAP-null cell pairs; pancreatic, lung, glioma lines among the validation panel
- **Design:** MTA accumulates in MTAP-null cells, partially inhibits PRMT5; reducing MAT2A lowers SAM and removes the methyl donor that PRMT5 needs to compete with MTA for binding. The MAT2A/PRMT5/RIOK1 axis is the proposed druggable node.
- **n:** screen across ~400 cell lines; isogenic validation in ~6 pairs
- **Effect size:** strong — Independent confirmation that MAT2A loss is selectively lethal in MTAP-deleted cells via PRMT5 substrate depletion. The paper coined the MAT2A/PRMT5/RIOK1 axis label that subsequent IDEAYA / Agios / Servier MAT2A programs used as their rationale.
- **Variance:** vs MTAP-WT isogenic; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Third of the three 2016 foundational papers. Marjon was at Agios at the time; this work led directly to AG-270 (later S095033) and the IDE397 / IDE892 IDEAYA program.

Raub TJ/De Dios A (2015) — Drug Metabolism and Disposition

Reference:PMID 26149830 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Abemaciclib brain penetration and orthotopic glioma activity
- **Indication / model:** Rat orthotopic U87MG GBM xenograft; rodent plasma + brain PK; in-vitro BBB transporter assays (P-gp, BCRP)
- **Design:** Abemaciclib is a weak P-gp/BCRP substrate, unlike palbociclib (which is a strong P-gp substrate), so unbound brain concentrations of abemaciclib reach pharmacologically active levels with normal BBB. The CDK4/6 inhibition itself blocks Rb phosphorylation and arrests CDKN2A-loss/Rb-intact tumors at the G1/S checkpoint.
- **n:** n=8-10 rats/arm in survival studies; transporter assays in replicate
- **Effect size:** moderate — Abemaciclib monotherapy at 40 and 80 mg/kg daily extended median survival by ~8 and 12 days vs vehicle in the rat orthotopic U87MG model; unbound brain exposures exceeded CDK4/6 Ki values. Palbociclib head-to-head reached brain levels too but at much higher doses and shorter dwell times. The paper is the proof that abemaciclib is the CNS-deployable CDK4/6 inhibitor.
- **Variance:** vs Vehicle; head-to-head with palbociclib in some comparisons; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** U87MG is a long-established but criticized GBM line — it has lost much of its parental-tumor signature

over decades of passage and is not strictly CDKN2A-null in all sublines. The PK and BBB data are robust regardless; the survival benefit is the part to read with the cell-line caveat. PDX validation came later in the Sahebjam / Wen INSIGHt preclinical lead-in.

Parrish KE/Elmquist WF (2015) — Molecular Cancer Therapeutics / J Pharmacol Exp Ther

Reference:PMID 26354993 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Palbociclib BBB efflux limits orthotopic activity (the Taal 2018 mechanism)
- **Indication / model:** Transgenic P-gp / BCRP knockout mice; orthotopic GBM22 PDX intracranial vs flank; PK in WT vs knockout brain
- **Design:** Palbociclib is a strong substrate for both P-gp and BCRP. Active efflux at the BBB strips drug from brain parenchyma — steady-state brain exposure was ~115-fold higher in transporter-deficient mice than in WT, and elacridar partially rescued WT brain levels.
- **n:** n=6-10 mice/arm across genotypes and tumor sites
- **Effect size:** negative — Palbociclib was efficacious against GBM22 implanted in the flank but failed against the same line implanted intracranially in WT mice; the failure tracked precisely to inadequate brain drug delivery via P-gp/BCRP efflux. This is the mechanistic explanation for the Taal 2018 clinical futility readout — wrong drug for the location, right pathway.
- **Variance:** vs WT mice; flank vs intracranial implantation matched; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The companion Brastianos / Vogelbaum work confirmed the same in independent models. The lesson — CDK4/6 inhibitor choice for CNS tumors is gated by BBB transporter pharmacology, not by potency at the kinase. Abemaciclib and ribociclib both clear; palbociclib does not.

Johnson LA/June CH (2015) — Science Translational Medicine

Reference:PMID 25696001 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Humanized EGFRvIII CAR-T validation (Penn — the construct used in O'Rourke 2017)
- **Indication / model:** Humanized scFv panel screened against EGFRvIII vs WT EGFR; NSG mouse with U87-EGFRvIII subcutaneous and orthotopic xenograft
- **Design:** The team generated a panel of humanized scFvs and selected a low-affinity binder specific for EGFRvIII over WT EGFR — the affinity tuning is what prevents on-target/off-tumor activity against normal EGFR-expressing skin, gut, and lung. Then incorporated into a second-generation CAR.
- **n:** Multiple scFv clones tested; n=8-10 mice/arm in xenograft validation
- **Effect size:** moderate — The humanized EGFRvIII CAR-T controlled xenogeneic human EGFRvIII+ glioblastoma in NSG mice and showed minimal cross-reactivity with WT EGFR-expressing tissues at clinically relevant doses. The construct went directly into the NCT02209376 phase 1 (Penn) that O'Rourke 2017 STM reported.
- **Variance:** vs Empty CAR or untransduced T cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** NSG mice have no functional immune system, so the tumor-microenvironment immunosuppression seen in the human trial (intratumoral PD-L1, IDO, Tregs in O'Rourke 2017) is absent in the model. The xenograft validation is necessary but not sufficient — the clinical activity was substantially lower than the preclinical signal would predict.

Reilly EB/Phillips AC (2015) — Molecular Cancer Therapeutics

Reference: [PMID 25731184](#) · Type: preclinical · Inclusion: included

- **Intervention:** Depatux-m (ABT-414, EGFR-directed ADC) preclinical
- **Indication / model:** EGFR-amplified and EGFRvIII+ GBM PDX panel; non-EGFR-amplified controls; xenograft and orthotopic
- **Design:** Depatux-m is an antibody-drug conjugate: the antibody portion (ABT-806) is selective for activated or amplified EGFR including EGFRvIII; the payload is monomethyl auristatin F (MMAF), a microtubule disruptor. EGFR-amplified tumors internalize the ADC, MMAF kills the cell, and bystander effects extend to nearby cells with lower EGFR.
- **n:** Multiple PDX lines; n=8-10 mice/arm
- **Effect size:** strong — Depatux-m produced regressions in EGFR-amplified GBM PDX models including EGFRvIII-positive lines. Activity was strictly EGFR-amplification-dependent — non-amplified controls did not respond. Synergy with temozolomide was observed in head-to-head xenograft studies, motivating the INTELLANCE-1 design.
- **Variance:** vs Isotype-control ADC; vehicle; head-to-head with temozolomide; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Strong preclinical signal, negative phase 3 (INTELLANCE-1, Lassman 2023). Classic example of preclinical-to-clinical mismatch — the ocular toxicity from MMAF and the heterogeneous EGFR expression in real human GBM both limited clinical effect. Logged here as the canonical preclinical-to-clinical caution for the EGFR-ADC class.

Sampson JH/Johnson LA (2014) — Clinical Cancer Research

Reference: [PMID 24352643](#) · Type: preclinical · Inclusion: included

- **Intervention:** EGFRvIII mCAR T-cell therapy cures intracerebral glioma (foundational preclinical)
- **Indication / model:** Fully immunocompetent C57BL/6 mouse with syngeneic SMA-560 / KR158 intracerebral glioma engineered to express EGFRvIII; third-generation EGFRvIII-specific murine CAR
- **Design:** Third-generation CAR (CD28 + 4-1BB) targeting EGFRvIII — a tumor-specific neoantigen with no normal-tissue expression. Engineered T cells traffic to the CNS, lyse EGFRvIII+ cells, and (the surprise finding) generate host immunity against EGFRvIII-negative tumor antigens via epitope spreading.
- **n:** n=8-10 mice/arm across multiple efficacy experiments
- **Effect size:** strong — At elevated doses, EGFRvIII mCAR T cells cured all mice with established intracerebral glioma. Cured mice were resistant to rechallenge with EGFRvIII-negative tumors, implying epitope spreading and durable host immunity. The paper is the preclinical foundation for the Penn EGFRvIII CAR-T (O'Rourke 2017 STM) and the entire EGFRvIII-directed cellular-therapy program.
- **Variance:** vs Untransduced T cells; irrelevant CAR; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Syngeneic mouse model with engineered EGFRvIII expression — homogeneous antigen expression, unlike human GBM where EGFRvIII is heterogeneous within tumor (Nathanson 2014). The dramatic cure rate did not translate fully to human trials, where antigen escape limited durability. Still the right preclinical proof of concept.

Nathanson DA/Mischel PS (2014) — Science

Reference:PMID 24310612 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EGFRvIII on extrachromosomal DNA — dynamic and reversible expression
- **Indication / model:** Patient-derived GBM cell lines, EGFRvIII+ at baseline; in-vitro EGFR TKI selection; FISH for chromosomal vs extrachromosomal EGFRvIII; xenograft validation
- **Design:** EGFRvIII is frequently encoded on extrachromosomal circular DNA (ecDNA), not on the chromosome. Under EGFR TKI pressure, EGFRvIII ecDNA is rapidly lost from the population; on TKI withdrawal, ecDNA EGFRvIII re-amplifies. Resistance is genetic (ecDNA loss) but reversible — the antigen comes back when the selective pressure ends.
- **n:** Multiple patient-derived lines; xenograft n=6-10/arm
- **Effect size:** strong — EGFRvIII ecDNA expression is highly dynamic. TKI exposure suppresses EGFRvIII; drug holiday re-amplifies it. Erlotinib resistance reverses within 72 h of drug withdrawal. The implication for CAR-T and vaccine therapy: antigen loss observed on therapy may be transient rather than durable, opening the door to sequential or pulsed antigen-directed strategies.
- **Variance:** vs Untreated lines; off-drug recovery cohort; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mostly in-vitro work; the in-vivo dynamics on a clinical timescale are less well-characterized. The paper changed how the field thinks about EGFRvIII heterogeneity — not a stable lineage trait but a population-level fluctuation.

Roth JC/Markert JM (2014) — Human Gene Therapy Clinical Development

Reference:PMID 24649838 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** M032 — oncolytic HSV-1 expressing IL-12 (preclinical)
- **Indication / model:** Aotus non-human primate intracerebral inoculation; mouse intracranial tumor models; biodistribution + neurovirulence safety
- **Design:** M032 is a γ 134.5-deleted HSV-1 carrying a transgene encoding murine or human IL-12. Tumor-restricted replication kills tumor cells directly; IL-12 secreted locally activates NK and T cells and shifts the tumor microenvironment toward a Th1 program.
- **n:** NHP n=3-6 per dose; mouse efficacy n=8-10/arm
- **Effect size:** moderate — M032 was non-pathogenic in Aotus NHPs across the full dose range and produced durable tumor regression in mouse intracranial models. IL-12 expression was confirmed in tumor tissue but not in normal brain — the local IL-12 payload is what differentiates M032 from G207 (its non-armed predecessor). The toxicology package supported the human phase 1 (NCT02062827).
- **Variance:** vs Parental HSV-1; vehicle; G207 head-to-head in some experiments; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Aotus is the standard primate model for HSV neurovirulence but is not a perfect human predictor. Mouse efficacy used implanted tumor lines, not GEMMs. M032 is conceptually attractive (IL-12 locally) but has not yet shown the clinical signal that CAN-3110 has — phase 1 readout pending.

Pyonteck SM/Joyce JA (2013) — Nature Medicine

Reference:PMID 24056773 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CSF-1R inhibition reshapes glioma myeloid microenvironment (TME context)

- **Indication / model:** PDGFB-driven RCAS GEMM glioma; CSF-1R inhibitor PLX3397; orthotopic transplant models
- **Design:** Tumor-associated macrophages (TAMs) in glioma express CSF-1R and are educated by glioma-derived CSF1 toward an M2-like, immunosuppressive phenotype. CSF-1R blockade does not deplete macrophages but reprograms them toward an antitumor program — reducing IL-10, TGF- β , and Arg1, increasing MHC-II and IFN- γ -responsive genes.
- **n:** n=8-12 mice/arm; multiple GEMM cohorts
- **Effect size:** strong — PLX3397 produced durable regressions in PDGFB-driven glioma GEMM by reprogramming TAMs rather than depleting them. Survival was extended substantially. The paper is the canonical demonstration that the myeloid compartment is the dominant immunosuppressive node in glioma — more so than T-cell-intrinsic checkpoints.
- **Variance:** vs Vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PDGFB-driven GEMM is a proneural-subtype model; the same logic may not hold for mesenchymal GBM. Clinical translation has been disappointing — the Butowski / Quail follow-up work showed adaptive PI3K reactivation in TAMs after long-term PLX3397, blunting durable benefit. The mechanism evidence is the part to carry forward: it is the strongest published reason why generic checkpoint inhibition fails in GBM, and why myeloid-targeted combinations are the next move.

Snuderl M/lafraite AJ (2011) — Cancer Cell

Reference:PMID 22137795 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EGFRvIII heterogeneity within GBM (the resistance biology)
- **Indication / model:** Multi-region tumor sampling from 18 primary EGFR-amplified GBMs; FISH and IHC for EGFRvIII; single-cell-level antigen mapping
- **Design:** EGFRvIII is expressed in a subset of cells within EGFR-amplified GBM, often in non-overlapping spatial clones with other EGFR variants (vIII, A289V, etc). The intratumoral heterogeneity means that any single-antigen-directed therapy leaves antigen-negative subclones to repopulate the tumor.
- **n:** 18 primary tumors; 3-5 regions each
- **Effect size:** strong — EGFRvIII positivity is heterogeneous within tumors and between regions in the same tumor — almost always co-existing with EGFRvIII-negative subclones in the same lesion. This is the underlying biology that explains the rindopepimut ACT IV failure and the antigen-loss escape seen in the O'Rourke 2017 CAR-T trial. Single-antigen targeting cannot work without addressing the negative subpopulation.
- **Variance:** vs Not applicable (descriptive); translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive heterogeneity study — no therapeutic intervention. The clinical implication is structural: any monovalent EGFRvIII therapy (vaccine, ADC, CAR-T) carries built-in resistance. Justifies the move toward dual-target (EGFR + IL13Ra2) and synNotch (E-SYNC) designs.

von Baumgarten L/Winkler F (2011) — Clinical Cancer Research

Reference:PMID 21788357 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Bevacizumab vascular and tumor-cell effects in GBM (dose-dependent)
- **Indication / model:** U87 and U251 GBM xenografts and orthotopic intracranial models; dose-ranging

anti-VEGF (bevacizumab)

- **Design:** Bevacizumab binds and neutralizes VEGF-A, blocking VEGF-receptor signaling on tumor endothelium. The effects are dose-dependent: low doses normalize vasculature (improving perfusion and possibly oxygenation); higher doses regress vessels and produce tumor cell apoptosis but compromise downstream drug delivery and increase hypoxia-driven invasive escape.
- **n:** n=8-10 mice/arm across dose levels
- **Effect size:** moderate — Medium-dose bevacizumab regressed glioma vasculature and tumor growth in orthotopic models, while higher doses produced more vascular destruction but did not further extend survival — the therapeutic window has an upper edge. Tumor-cell-intrinsic effects (VEGFR signaling on tumor cells) accounted for part of the activity, not only vascular normalization.
- **Variance:** vs Vehicle / IgG isotype; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Xenograft, not GEMM. The clinical bevacizumab story (no OS benefit in newly diagnosed GBM, modest PFS gain in recurrent) is consistent with the preclinical hint that vascular regression alone is insufficient. The intracranial hemorrhage caveat in this case (patient presented with intratumoral hemorrhage) is not directly tested here but is implicit in the vascular-disruption mechanism.

Ostertag D/Robbins JM (2011) — Neuro-Oncology

Reference:PMID 22070930 · **Type:** preclinical · **Inclusion:** excluded — Toca 511 program is discontinued (TOCA 5 phase 3 was negative; sponsor Tocagen wound down). The preclinical signal was strong (durable CR in orthotopic models) but the clinical translation failed and the drug is not enrollable. Logged as audit trail for the oncolytic / gene-therapy axis; relevant context for why CAN-3110 and PVSRIPO are the current active assets, not Toca 511.

- **Intervention:** Toca 511 retroviral cytosine deaminase + 5-FC preclinical
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Toca 511 program is discontinued (TOCA 5 phase 3 was negative; sponsor Tocagen wound down). The preclinical signal was strong (durable CR in orthotopic models) but the clinical translation failed and the drug is not enrollable. Logged as audit trail for the oncolytic / gene-therapy axis; relevant context for why CAN-3110 and PVSRIPO are the current active assets, not Toca 511.

Michaud K/Stokoe D (2010) — Cancer Research

Reference:PMID 20354191 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PD-0332991 (palbociclib) in INK4a-ARF-null orthotopic glioma
- **Indication / model:** U87MG (INK4a-ARF-null) and GBM39 intracranial xenograft; flank xenograft for comparison
- **Design:** INK4a/ARF (CDKN2A) loss derepresses CDK4/6, which phosphorylates Rb and drives G1/S transition. CDK4/6 inhibition reimposes the checkpoint in Rb-intact tumors.
- **n:** n=8-10 mice/arm
- **Effect size:** moderate — Palbociclib arrested growth of INK4a-ARF-null orthotopic U87MG xenografts and extended survival modestly. The paper is one of the first proofs that CDK4/6 inhibition has activity against CDKN2A-null intracranial glioma — though the orthotopic effect was substantially weaker than the flank effect, foreshadowing the BBB-efflux explanation that Parrish 2015 nailed down.
- **Variance:** vs Vehicle; translatability: med
- **Toxicities:** n/a (preclinical)

- **Case fit:** partial
- **Notes:** Early-era study — uses U87MG, predates the BBB-efflux literature. The signal is real but weaker than expected from the genetics, which Parrish 2015 later explained mechanistically.

Sampson JH/Bigner DD (2010) — Journal of Clinical Oncology

Reference: [PMID 20921459](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Rindopepimut — phase 2 with immune-escape mechanism (ACTIVATE / ACT II)
- **Indication / model:** EGFRvIII+ newly diagnosed GBM patients (n=18 ACTIVATE) with paired pre/post-vaccine tumor sampling; mouse intracerebral SMA-560 efficacy backdrop
- **Design:** Synthetic 14-mer peptide spanning the EGFRvIII fusion junction conjugated to KLH, given with GM-CSF adjuvant. Generates EGFRvIII-specific antibodies and T cells. Mechanism of failure: vaccinated patients lost EGFRvIII expression at recurrence, demonstrating immune-driven antigen escape in human tumors.
- **n:** n=18 vaccinated patients with paired tumor sampling at recurrence
- **Effect size:** moderate — EGFRvIII expression was lost in 82% of recurrent tumors from vaccinated patients vs persistent expression in untreated controls. The paper coined the 'immunologic escape' framing for EGFRvIII vaccines. mPFS ~14.2 mo and mOS ~26 mo were encouraging in single-arm context; phase 3 ACT IV later showed the OS benefit did not survive blinded comparison — heterogeneous EGFRvIII allows antigen-negative subclones to escape.
- **Variance:** vs Pre-vaccine baseline (paired); translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Logged primarily as a board-relevant caution: strong preclinical signal + positive phase 2 + negative phase 3. The antigen-loss biology (Snuderl 2011, Nathanson 2014) was the proximate failure mechanism. The same lesson informs the patient's EGFRvIII CAR-T trial selection — dual-target and synNotch designs are the post-rindopepimut response.

Stupp R/Mirimanoff RO (2009) — Lancet Oncology

Reference: [PMID 19269895](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Temozolomide + radiotherapy (5-year follow-up)
- **Indication / model:** Newly diagnosed glioblastoma — long-term follow-up (1L); Full EORTC 26981 cohort with 5+ years of follow-up.
- **Design:** Long-term follow-up of EORTC 26981 RCT
- **n:** 573
- **Effect size:** 5-yr OS 9.8% (TMZ+RT) vs 1.9% (RT) % — OS at 2, 3, 4, 5 years
- **Variance:** p=<0.0001
- **Toxicities:** No new late toxicities beyond the parent trial. MGMT methylation was the dominant predictor of long-term survival; methylated MGMT and TMZ exposure together identified the patients who reached 5-year survival.
- **Case fit:** strong
- **Notes:** 5-year follow-up confirms that the OS benefit holds out and that long-term survivors are enriched for MGMT methylation. Late-AE reporting was minimal in this update; the parent Stupp 2005 paper carries the index toxicity table.

Friedman HS/Cloughesy TF (2009) — Journal of Clinical Oncology

Reference:PMID 19720927 · Type: clinical · Inclusion: included

- **Intervention:** Bevacizumab (BRAIN study, recurrent GBM)
- **Indication / model:** Recurrent glioblastoma (2L+); 167 adults with recurrent GBM (first or second relapse), KPS ≥ 70 . Randomized 1:1 to bevacizumab monotherapy or bevacizumab + irinotecan.
- **Design:** Multicenter open-label phase 2 (non-comparative)
- **n:** 167
- **Effect size:** 42.6 % (PFS6) — PFS6, ORR
- **Variance:** 95% CI 29.6–55.5
- **Toxicities:** intracranial hemorrhage $G \geq 3$ (n=167, 2.4%); hypertension $G \geq 3$; proteinuria; venous thromboembolism $G \geq 3$; wound dehiscence; treatment discontinuation due to AE
- **Case fit:** partial
- **Notes:** BRAIN study supported FDA accelerated approval of bevacizumab for recurrent GBM. CRITICAL safety flag in this case: the patient presented with intratumoral hemorrhage, which is the most cited relative contraindication for bevacizumab. Carry forward as a recurrence option only after explicit re-imaging and bleeding-risk reassessment with the treating team.

Stupp R/Mirimanoff RO (2005) — N Engl J Med

Reference:PMID 15758010 · Type: clinical · Inclusion: included

- **Intervention:** Temozolomide + radiotherapy (Stupp regimen)
- **Indication / model:** Newly diagnosed glioblastoma (1L); Adults age 18-70 with newly diagnosed histologically confirmed GBM, WHO PS 0-2, post-resection or biopsy. MGMT status not used for stratification at randomization but later shown to be the strongest effect modifier.
- **Design:** Randomized phase 3 (EORTC 26981-22981 / NCIC CE.3)
- **n:** 573
- **Effect size:** 0.63 HR — OS
- **Variance:** 95% CI 0.52–0.75; $p < 0.001$
- **Toxicities:** grade 3-4 hematologic AE during concurrent phase $G \geq 3$ (n=284, 7%); grade 3-4 hematologic AE during adjuvant phase $G \geq 3$ (n=223, 14%); any grade 3-4 AE $G \geq 3$; pneumocystis pneumonia
- **Case fit:** strong
- **Notes:** The trial that defined modern GBM standard of care. Patient is 66 — at the upper edge of the original 18-70 age inclusion. The OS benefit holds across age subgroups but is largest in MGMT-methylated tumors (see Hegi 2005 companion paper, PMID 15758009).

Hegi ME/Stupp R (2005) — N Engl J Med

Reference:PMID 15758009 · Type: clinical · Inclusion: included

- **Intervention:** Temozolomide + radiotherapy stratified by MGMT methylation
- **Indication / model:** Newly diagnosed glioblastoma; MGMT-methylated vs unmethylated subsets (1L); MGMT promoter methylation by methylation-specific PCR on 206 of 573 EORTC 26981 patients with assessable tumor. Methylated 45%, unmethylated 55%.
- **Design:** Post-hoc biomarker analysis of EORTC 26981
- **n:** 206

- **Effect size:** Methylated mOS 21.7 vs 15.3 mo (TMZ+RT vs RT); unmethylated mOS 12.7 vs 11.8 mo months — OS by MGMT methylation status
- **Variance:** $p=0.007$ (methylated subset)
- **Toxicities:** No new safety signals; this is a biomarker analysis of EORTC 26981, safety reported in the parent trial.
- **Case fit:** strong
- **Notes:** The landmark biomarker paper that established MGMT promoter methylation as the dominant predictor of TMZ benefit in GBM. Critical for this case because the patient's MGMT status is unknown — methylation will dictate whether the adjuvant TMZ commitment is high-yield or marginal. Toxicity profile is reported in the Stupp 2005 parent trial; this paper does not break out AE rates by MGMT subgroup.